

One-Phase Euler Coordinates for Affine Brownian Factors

Exact classification, phase diffusion, and conjugate normal-model pricing

Complex–Itô Random-Walk Research Project

July 25, 2026

We classify the nondegenerate affine complex Brownian processes $Z_t = Z_0 + at + bW_t$ that admit an exact, lossless, time-independent Euler representation $Z_t = Z_0 r(\theta_t) e^{i\theta_t}$ in which the genuine polar phase θ_t is the only stochastic state coordinate. For $Z_0 b \neq 0$, such a representation exists if and only if $a/b \in \mathbb{R}$ and $b/Z_0 \notin \mathbb{R}$. The first condition fixes all temporal supports to one affine line; the second makes its argument injective by requiring that line to miss the polar origin. Writing $a = \lambda b$ and $b/Z_0 = \rho e^{i\phi}$, the representation and inverse are $r(\theta) = \sin \phi / \sin(\phi - \theta)$ and $x(\theta) = \sin \theta / [\rho \sin(\phi - \theta)]$ on a unique open phase interval of length π . The canonical financial chart is $1 + iX_t/c = \sec \theta_t e^{i\theta_t}$ with $\theta_t = \arctan(X_t/c)$. We derive its Itô and Stratonovich equations, generator, exact transition kernel, natural boundaries, and transported additive group. A rigidity result shows that constant-coefficient additive dynamics force a line, whereas compatible multiplicative dynamics force a logarithmic spiral. Under a pricing measure, the chart conjugates the Bachelier generator, option values, Greeks, and hedges without changing the financial model. Reproducible experiments verify reconstruction, strong convergence, transition and pricing PDEs, and exact Bachelier prices and Greeks. Matched controls show that the chart creates neither an additional stochastic dimension nor automatic Monte Carlo variance reduction, holomorphic structure, path compression, or universal numerical acceleration. The result is therefore an exact geometric coordinate and a classification theorem, not a replacement for Itô calculus or a new pricing model.

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1 Introduction

Euler’s identity

$$e^{i\theta} = \cos \theta + i \sin \theta$$

turns phase into planar rotation, while the Itô table records the quadratic variation of Brownian motion through $d[W, W]_t = dt$. Their superficial resemblance—each contains a consequential square—often motivates an attempt to combine drift and Brownian noise into one complex exponential. The motivation is geometrically natural, but the two squares have different meanings. The identity $i^2 = -1$ is algebraic. Brownian quadratic variation is a limiting path property. A valid bridge between them must pass through Itô’s formula, support geometry, and an invertible state map.

This paper asks a precise version of that question. Given a real standard Brownian motion W and

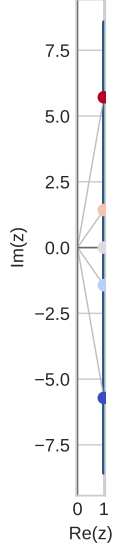
$$Z_t = Z_0 + at + bW_t, \quad Z_0, a, b \in \mathbb{C}, \quad (1)$$

when can the complete process be written

$$Z_t = Z_0 r(\theta_t) e^{i\theta_t} \quad (2)$$

with a deterministic, time-independent radius r and one real stochastic phase θ that is both the genuine polar argument and a lossless state coordinate? The requirement is substantially stronger than merely encoding a real variable by some measurable real number. It asks the polar angle itself to carry the random state while the modulus is a deterministic function of that angle.

The secant polar graph is a straight affine line



Radius diverges at inaccessible endpoints

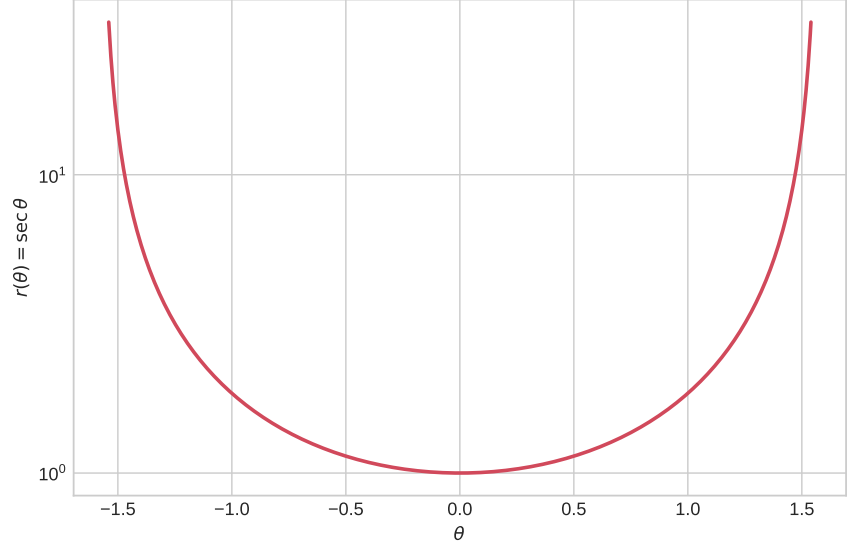


Figure 1: The canonical secant polar graph, its affine-line image, and the radius divergence at the inaccessible phase endpoints.

The central result is an exact classification. In the nondegenerate case $Z_0 b \neq 0$, Eq. 2 exists if and only if

$$\boxed{\frac{a}{b} \in \mathbb{R}, \quad \frac{b}{Z_0} \notin \mathbb{R}.} \quad (3)$$

The first condition says that drift and diffusion are real-collinear, so all fixed-time supports coincide with one affine line. The second says that the line misses the origin, making its continuous argument strictly monotone. If

$$a = \lambda b, \quad \frac{b}{Z_0} = \rho e^{i\phi}, \quad x_t = \lambda t + W_t,$$

then the exact formula is

$$\boxed{Z_t = Z_0 \frac{\sin \phi}{\sin(\phi - \theta_t)} e^{i\theta_t}, \quad \theta_t = \arg(1 + \rho e^{i\phi} x_t).} \quad (4)$$

The inverse is explicit and the admissible phase interval has length π . Thus an additive affine process can acquire a genuine, lossless stochastic phase, but it cannot execute unrestricted winding.

The canonical form is especially transparent. For an arithmetic Brownian factor X and scale $c > 0$,

$$\boxed{1 + i \frac{X_t}{c} = \sec \theta_t e^{i\theta_t}, \quad \theta_t = \arctan\left(\frac{X_t}{c}\right), \quad X_t = c \tan \theta_t.} \quad (5)$$

The state formula contains one stochastic variable, θ_t . Time still indexes the process and Brownian randomness still drives its law; neither can be removed from a nontrivial stochastic model. The point is state factorization, not the disappearance of stochasticity.

Figure 1 reveals the principal geometric fact. Although $\sec \theta$ and $e^{i\theta}$ look radial and rotational separately, their product is

$$\sec \theta e^{i\theta} = 1 + i \tan \theta,$$

a straight line. The representation changes coordinates and traversal speed; it does not turn rank-one additive Brownian motion into planar Brownian motion.

1.1 Contributions

The paper makes five contributions.

1. It distinguishes measurable scalar factorization from a genuine, continuous, lossless Euler phase and proves the complete affine classification Eq. 3, including a randomized-time necessity argument that respects almost-sure quantifiers.
2. It derives the general phase radius, inverse, domain, Itô and Stratonovich dynamics, generator, transition density, semigroup, exact step, and boundary classification.
3. It proves a local rigidity dictionary: constant additive complex noise forces a straight line, whereas compatible constant multiplicative noise forces a logarithmic spiral. It also identifies the transported group law of the phase chart.
4. It develops a financial interpretation in which the complex state is an auxiliary coordinate for a real Bachelier factor. Pricing, Greeks, hedging, and the pricing PDE are conjugate under the chart.
5. It subjects every broad implication to matched numerical controls. Those controls separate exact coordinate benefits from unsupported claims of added dimension, winding, variance reduction, path compression, holomorphic calculus, or universal acceleration.

The arctangent transform, stochastic changes of variables, Bachelier pricing, compactification of unbounded domains, stochastic exponentials, and complex linear SDEs are established subjects. The contribution is the classification and the integrated stochastic-financial analysis, not the invention of those ingredients.

1.2 Organization

Section 2 positions the result. Section 3 fixes the probability, geometric, and financial assumptions. Section 4 isolates measurable factorization and support geometry. Section 5 proves the affine theorem and handles exceptional cases. Section 6 develops the canonical secant chart, followed by its stochastic theory in Section 7. Section 8 compares additive lines, multiplicative spirals, and transported groups. Section 9 establishes pricing conjugacy and Section 10 reports the reproducible evidence. Section 11 and Section 12 delimit the vector and complex-analytic implications. Section 13 gives qualified applications and open problems. Technical derivations and reproducibility details appear in the appendices.

2 Literature and novelty boundary

2.1 Brownian motion, random walks, and finance

Bachelier’s 1900 dissertation introduced a Gaussian diffusion model for speculative prices and an early option-pricing calculation (Bachelier 1900). Wiener’s path-space construction provided a mathematical foundation for Brownian motion (Wiener 1923), and Donsker’s invariance principle later made the random-walk-to-Brownian limit precise (Donsker 1951). Modern Brownian-motion treatments supply the support, path, and quadratic-variation facts used below (Mörters and Peres 2010). We do not retroactively attribute modern semimartingale or risk-neutral machinery to Bachelier; those frameworks came later.

Itô’s foundational papers introduced stochastic integration and the second-order change-of-variables rule (Itô 1944, 1951). Girsanov established absolutely continuous measure changes for diffusion processes (Girsanov 1960), and Kac connected Wiener functionals to parabolic equations (Kac 1949). Modern derivative valuation rests on no-arbitrage, replication, and martingale arguments developed in particular by Black and Scholes, Merton, and Harrison and Pliska (Black and Scholes 1973; Merton 1973; Harrison and Pliska 1981).

2.2 Transformations and angular coordinates

Nonlinear diffusion coordinates are classical. Lamperti transformations normalize scalar diffusion coefficients (Lamperti 1964). Doss and Sussmann relate suitable stochastic equations to pathwise ordinary equations (Doss 1977; Sussmann 1977), while Zvonkin uses a one-to-one phase-space transformation to remove irregular drift (Zvonkin 1974). Särkkä and Solin use an arctangent transform of Brownian motion as an exact nonlinear-SDE benchmark (Särkkä and Solin 2019). These works rule out any novelty claim for applying arctan or Itô’s formula to a scalar diffusion.

Projecting a Gaussian vector onto an angle is also established in directional statistics (Pukkila and Rao 1988; Wang and Gelfand 2013). The law studied here is different from a generic projected bivariate normal: its Gaussian support is one affine line, and its phase density follows from a one-dimensional monotone change of variables.

Genuine planar Brownian winding requires two-dimensional motion and has asymptotic behavior absent from a one-line process (Spitzer 1958; Pitman and Yor 1986). Stereographic projection of Brownian motion likewise has a substantial prior theory (Carne 1985). Those literatures are comparisons, not derivations of the present phase.

2.3 Multiplicative and complex stochastic models

Stochastic exponentials and logarithms are part of established semimartingale calculus (Doléans-Dade 1970; Larsson and Ruf 2019). Modern representation and multiplicative-compensation frameworks further systematize this theory (Černý and Ruf 2022, 2023). Complex scalar linear equations and complex geometric Brownian motion appear explicitly in numerical-stability research (Buckwar et al. 2006; Perret 2010). Accordingly, the logarithmic-spiral comparator derived below is not a new complex GBM. Its role is to expose why unbounded lossless winding is compatible with multiplicative, but not additive, constant-coefficient dynamics.

2.4 Financial and computational precedents

The normal model remains relevant when negative forwards or strikes are plausible. Schachermayer and Teichmann compare Bachelier and Black–Merton–Scholes asymptotics (Schachermayer and Teichmann 2008), while Choi et al. survey normal-model pricing, volatility conversion, and risk management (Choi et al. 2022). CME’s 2020 clearing advisory documents a practical switch to Bachelier valuation for specified energy options after negative prices (CME Group 2020).

Arctangent compactification is not new in financial PDEs. Duffie and Kan use an arctan coordinate in term-structure derivative calculations (p. 396) (Duffie and Kan 1996), and in ’t Hout and Snoeijer map $\mathbb{R}^d$ into a cube with componentwise arctangent coordinates (Hout and Snoeijer 2021). More broadly, mapped methods on unbounded domains have long been studied (Grosch and Orszag 1977; Boyd 1987), with option-pricing localization and high-dimensional approximations requiring problem-specific error analysis (Cont and Voltchkova 2005; Reisinger and Wittum 2007).

Higham provides standard SDE convergence methodology (Higham 2001). Exact diffusion simulation has a more demanding algorithmic meaning than applying a known inverse map (Beskos and Roberts 2005), and geometric or symmetry-adapted stochastic integrators constitute separate mature research directions

(Malham and Wiese 2008; De Vecchi et al. 2019). Monte Carlo and low-discrepancy methods entered finance through work including Boyle and Paskov–Traub (Boyle 1977; Paskov and Traub 1995). Early-exercise, path-dependent, and high-dimensional problems need information beyond a terminal one-factor coordinate (Longstaff and Schwartz 2001; Zhang 2003).

Finally, Wirtinger calculus is established for nonholomorphic complex objectives (Brandwood 1983), and complex-step differentiation has specific analytic-extension hypotheses (Squire and Trapp 1998). Complex methods already contribute to finance through characteristic functions and Fourier inversion, as in Heston and Carr–Madan (Heston 1993; Carr and Madan 1999). Those uses of complex analysis are conceptually distinct from the state chart developed here.

2.5 Claim boundary

Table 1: Novelty and non-novelty boundary.

Topic	Established status	Claim made here
$\arctan(X/c)$ diffusion	Known smooth transform and numerical benchmark	Canonical representative of the classified affine family
Euler and polar form	Elementary complex analysis	Exact coordinate language, not a new algebra
Angular Gaussian laws	Established directional statistics	Explicit line-supported specialization
Arctangent PDE compactification	Established numerical technique	Exact operator/pricing conjugacy and controlled benchmark
Stochastic exponential and complex GBM	Established	Multiplicative rigidity comparator
Bachelier valuation	Established	Financial verification and coordinate-invariance result
Affine iff-classification	No directly matching theorem found in the reviewed literature	Central theorem with necessity and sufficiency proof
Additive-line versus multiplicative-spiral rigidity	Components are classical	Unified coefficient-matching classification
Automatic variance reduction or new holomorphic calculus	False in general	Negative results and explicit controls

3 Probabilistic, geometric, and financial setup

3.1 Probability space and complex brackets

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ satisfy the usual conditions, and let W be a one-dimensional standard real Brownian motion. For a partition $\Pi = \{0 = t_0 < \dots < t_n = t\}$,

$$\sum_{j=0}^{n-1} (W_{t_{j+1}} - W_{t_j})^2 \longrightarrow t$$

in probability as the mesh tends to zero. Differentially,

$$d[W, W]_t = dt. \tag{6}$$

For a finite interval of length h , $\Delta W = \sqrt{h}\xi$ with $\xi \sim N(0, 1)$, so $(\Delta W)^2 = h\xi^2$, not h pathwise. Hence Eq. 6 is not the ordinary algebraic identity $i^2 = -1$.

Identify $\mathbb{C}$ with $\mathbb{R}^2$. If $Z = X + iY$ is a continuous complex semimartingale, retain both

$$[Z, Z] = [X] - [Y] + 2i[X, Y], \quad [Z, \overline{Z}] = [X] + [Y]. \quad (7)$$

For $dZ = b dW$,

$$d[Z, Z]_t = b^2 dt, \quad d[Z, \overline{Z}]_t = |b|^2 dt.$$

The complex coefficient records a scale and a planar direction. It does not create a second Brownian driver.

3.2 Genuine one-phase representation

Definition 3.1 (Genuine lossless one-phase representation). Let $I \subset \mathbb{R}$ be an interval. A process Z has a genuine, lossless, time-independent one-phase Euler representation relative to $Z_0 \neq 0$ if there are:

1. a deterministic Borel function $r : I \rightarrow (0, \infty)$;
2. a real adapted process θ taking values in I ; and
3. a deterministic inverse recovering the scalar affine state from θ_t ,

such that

$$Z_t = Z_0 r(\theta_t) e^{i\theta_t}$$

pathwise, θ_t is a continuous genuine argument lift of Z_t/Z_0 , and the same r works for all deterministic times in a nontrivial interval. When an SDE is derived directly, the state maps are additionally assumed C^2 .

“Genuine” excludes arbitrary state labels called angles: modulo 2π , θ must equal the geometric argument. “Lossless” excludes wrapped encodings that forget a winding integer or radial state. “Time-independent” excludes a moving polar graph $r(t, \theta)$. Equalities below are identified as pathwise, almost sure, distributional, or numerical as needed.

The smoothness distinction is essential. A Borel factor may define a random variable but cannot automatically be inserted into Itô’s formula. Conversely, a smooth one-to-one state chart is both measurable and analytically useful.

3.3 Financial interpretation

Financial claims are made under a separate pricing setup. Let $\mathbb{Q}$ be an equivalent martingale measure associated with a chosen numeraire. Let X be a real factor with

$$dX_t = \mu_{\mathbb{Q}}(t, X_t) dt + \sigma(t, X_t) dW_t^{\mathbb{Q}},$$

and assume existence, integrability, and payoff conditions sufficient for the stated conditional expectations and PDEs. When $X = F$ is a forward under its forward measure, $\mu_{\mathbb{Q}} = 0$ in the elementary normal model. The complex variable

$$\mathcal{Z} = 1 + iX/c$$

is an auxiliary coordinate, not a traded asset and not a claim that market prices are intrinsically complex. Discount factors, numeraires, and self-financing strategies retain their ordinary real financial meaning.

4 Measurable factorization and support geometry

Before imposing polar geometry, one should ask whether a complex state can factor through a scalar at all.

Theorem 4.1 (Measurable phase-radius criterion). *Let E be a standard Borel space, let $f : E \rightarrow \mathbb{C} \setminus \{0\}$ be Borel, and let $\vartheta : E \rightarrow \mathbb{R}$ be a Borel genuine argument lift. A Borel function r satisfies*

$$f(s) = r(\vartheta(s))e^{i\vartheta(s)}$$

if and only if $|f|$ is measurable with respect to $\sigma(\vartheta)$.

Proof. Necessity follows from $|f| = r \circ \vartheta$. Conversely, the Doob–Dynkin factorization principle gives a Borel r with $|f| = r \circ \vartheta$; combining this with the genuine-argument identity proves the claim (Kallenberg 2002). $\square$

A convenient fiber condition is

$$\vartheta(s_1) = \vartheta(s_2) \implies |f(s_1)| = |f(s_2)|. \quad (8)$$

If ϑ is Borel and injective, its inverse on its image is Borel, so Eq. 8 is sufficient for a lossless representation. Injectivity is not required for radius factorization alone, but it is required when the original scalar state must be recovered.

4.1 Polar graphs are planar-null

Lemma 4.1 (Borel polar graphs have zero planar measure). *For Borel $r : \mathbb{R} \rightarrow [0, \infty)$, the set*

$$C_r = \{r(\theta)e^{i\theta} : \theta \in \mathbb{R}\}$$

has two-dimensional Lebesgue measure zero.

Proof. The Borel image is analytic and Lebesgue measurable. For a fixed geometric angle $\psi \in [0, 2\pi)$, the positive ray at ψ meets C_r only in the countable set

$$\{r(\psi + 2\pi k)e^{i\psi} : k \in \mathbb{Z}\}.$$

The radial section therefore has one-dimensional measure zero. Polar Fubini integration gives

$$\lambda_2(C_r) = \int_0^{2\pi} \int_0^\infty \mathbf{1}_{C_r}(qe^{i\psi})q \, dq \, d\psi = 0. \quad \square$$

This elementary lemma is the key obstruction when a family of fixed-time support lines sweeps a two-dimensional region.

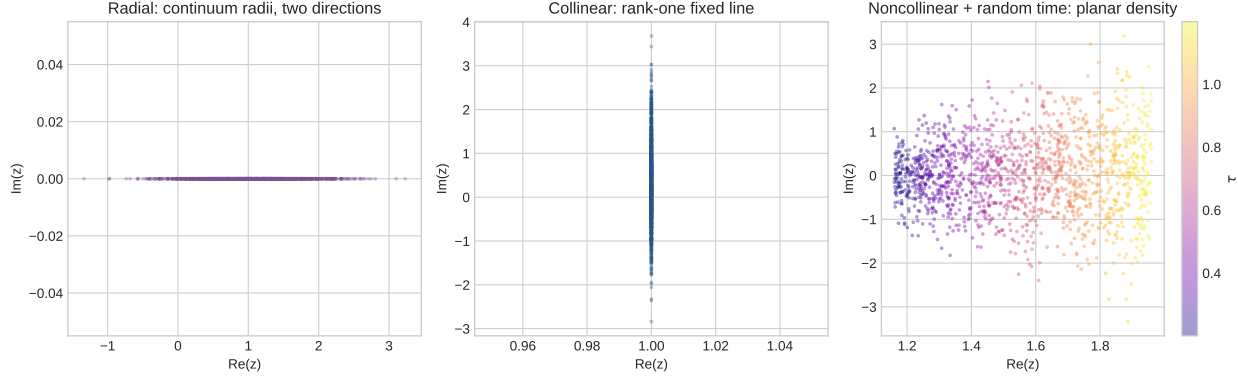


Figure 2: Radial, eligible collinear, and randomized-time noncollinear support geometries.

4.2 The three support geometries

For Eq. 1 with $b \neq 0$, fixed-time support is

$$\text{supp}(Z_t) = Z_0 + at + b\mathbb{R}. \quad (9)$$

There are three qualitatively different cases.

1. If $a/b \in \mathbb{R}$ and $b/Z_0 \notin \mathbb{R}$, all supports coincide with a nonradial affine line missing the origin. Its angle is monotone and lossless.
2. If $a/b \in \mathbb{R}$ and $b/Z_0 \in \mathbb{R}$, the fixed line passes through the origin. Its geometric angle supplies only two ray directions for a continuum of Gaussian radii.
3. If $a/b \notin \mathbb{R}$, the support line moves transversely with time. Randomizing time produces a planar law, which cannot live on one Borel polar graph.

The covariance-rank contrast in Figure 2 is illustrative, not the proof. In the reproducible experiment, the radial and eligible lines had empirical rank one, while the randomized-time noncollinear law had positive small eigenvalue 0.0531 and determinant 0.0370.

Figure 3 provides a second diagnostic. As angular bins are refined, conditional radial dispersion decays for a line on which radius is a function of angle; it remains substantial for a planar distribution. The diagnostic visualizes Eq. 8 and does not replace Lemma 4.1.

5 Exact affine classification

Normalize the process by Z_0 :

$$\zeta_t := \frac{Z_t}{Z_0} = 1 + \alpha t + cW_t, \quad \alpha = \frac{a}{Z_0}, \quad c = \frac{b}{Z_0}. \quad (10)$$

Theorem 5.1 (Lossless time-independent affine classification). *Assume $Z_0 b \neq 0$ and require Definition Definition 3.1 on every deterministic time in a nontrivial interval. Then*

$$Z_t = Z_0 r(\theta_t) e^{i\theta_t}$$

exists if and only if

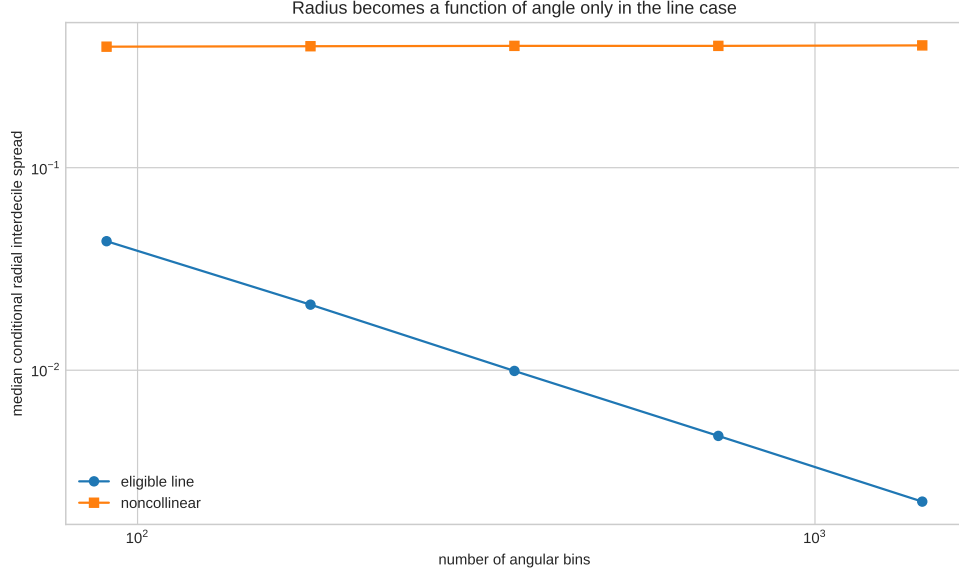


Figure 3: Conditional radial spread as angular bins are refined for an eligible line and a noncollinear randomized-time law.

$$\boxed{\frac{a}{b} \in \mathbb{R} \quad \text{and} \quad \frac{b}{Z_0} \notin \mathbb{R}.} \quad (11)$$

Equivalently,

$$\text{Im}(a\bar{b}) = 0, \quad \text{Im}(b\bar{Z}_0) \neq 0.$$

5.1 Sufficiency and explicit construction

Assume $a = \lambda b$ with $\lambda \in \mathbb{R}$, and write

$$c = \frac{b}{Z_0} = \rho e^{i\phi}, \quad \rho > 0, \quad \sin \phi \neq 0.$$

Then

$$\zeta_t = 1 + cx_t, \quad x_t = \lambda t + W_t. \quad (12)$$

The line $L = \{1 + cx : x \in \mathbb{R}\}$ misses the origin. Its continuous argument $\theta(x)$, anchored by $\theta(0) = 0$, satisfies

$$\frac{d\theta}{dx} = \text{Im} \frac{c}{1 + cx} = \frac{\text{Im } c}{|1 + cx|^2}, \quad (13)$$

which has fixed nonzero sign. Hence θ is strictly monotone and lossless.

Write $1 + \rho e^{i\phi} x = r e^{i\theta}$. Multiplication by $e^{-i\phi}$ and comparison of imaginary parts give

$$r \sin(\theta - \phi) = -\sin \phi.$$

Therefore

$$\boxed{r(\theta) = \frac{\sin \phi}{\sin(\phi - \theta)}, \quad x(\theta) = \frac{\sin \theta}{\rho \sin(\phi - \theta)}. \quad (14)}$$

Substitution into Eq. 12 proves the representation and its inverse.

5.2 Necessity of drift-diffusion collinearity

Suppose instead that $\text{Im}(a\bar{b}) \neq 0$. Choose an independent random time τ uniformly distributed on $[t_0, t_1] \subset (0, \infty)$. The pair (τ, W_τ) has density

$$f_{\tau, W_\tau}(t, w) = \frac{\mathbf{1}_{[t_0, t_1]}(t)}{t_1 - t_0} \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{w^2}{2t}\right). \quad (15)$$

The real-affine map

$$(t, w) \mapsto Z_0 + at + bw$$

has nonzero real determinant, equal up to orientation to $\text{Im}(a\bar{b})$. Thus Z_τ is absolutely continuous with respect to planar Lebesgue measure.

If one time-independent polar graph represented Z_t almost surely for every deterministic $t \in [t_0, t_1]$, Fubini's theorem would give $\mathbb{P}(Z_\tau \in Z_0 C_r) = 1$. By Lemma 4.1 and planar absolute continuity, the same probability must be zero, a contradiction. Therefore $a/b \in \mathbb{R}$.

Randomizing time is not cosmetic. It closes the quantifier gap in an informal argument that an uncountable family of support lines “fills an area”; an uncountable union of null events cannot otherwise be handled by countable probability arguments.

5.3 Necessity that the fixed line miss the origin

Under $a/b \in \mathbb{R}$, all supports coincide with $L = Z_0 + b\mathbb{R}$. If $b/Z_0 \in \mathbb{R}$, the normalized line passes through the origin. Every nonzero point then has geometric argument 0 or π modulo 2π . Any genuine unwrapped phase belongs to the countable set

$$\{2\pi k : k \in \mathbb{Z}\} \cup \{\pi + 2\pi k : k \in \mathbb{Z}\}.$$

A single-valued r maps this set to only countably many points. At every $t > 0$, however, Z_t has a nonatomic Gaussian law along the entire line. That law cannot be countably supported. Hence $b/Z_0 \notin \mathbb{R}$. Together with the construction above, this proves Theorem 5.1. $\square$

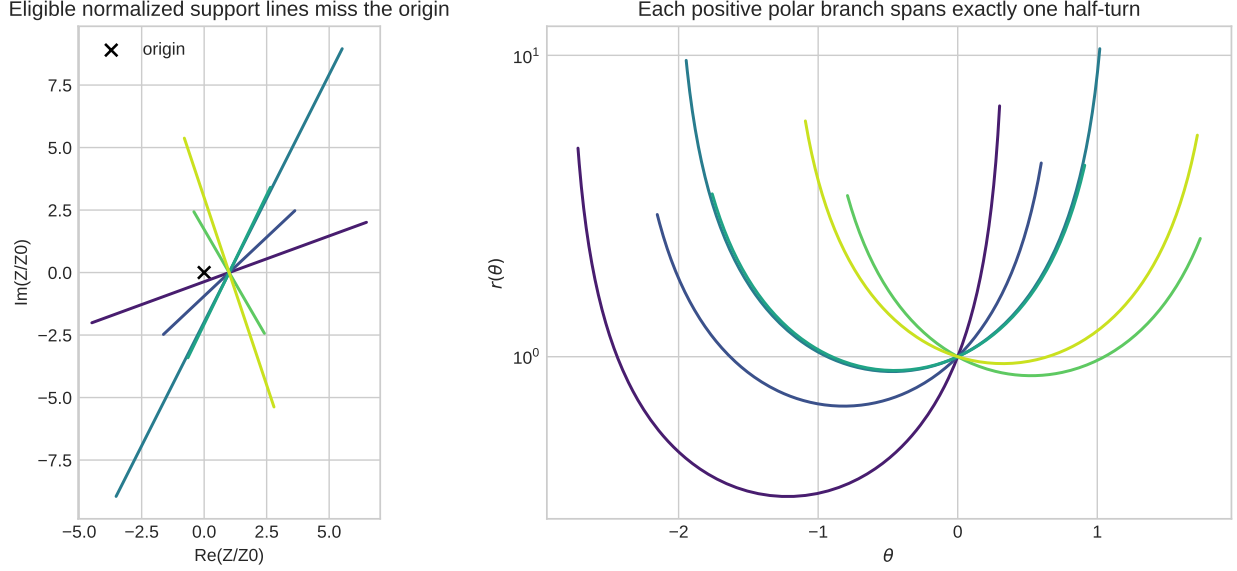


Figure 4: Six eligible oblique affine lines and their positive phase-radius branches.

5.4 Phase domain and uniqueness

The exact phase interval is

$$I_\phi = \begin{cases} (\phi - \pi, \phi), & \sin \phi > 0, \\ (\phi, \phi + \pi), & \sin \phi < 0. \end{cases} \quad (16)$$

It is the unique open interval of length π containing zero on which Eq. 14 has positive radius. The map $x \mapsto \theta$ is a bijection from $\mathbb{R}$ to I_ϕ , and

$$|x(\theta)| \rightarrow \infty, \quad r(\theta) \rightarrow \infty$$

at its endpoints. A continuous genuine argument lift is unique after $\theta(0) = 0$ is fixed. Discontinuous additions of state-dependent multiples of 2π are artificial encodings and do not define the continuous diffusion coordinate studied here.

The random-parameter reconstruction in Figure 4 reached maximum scalar inverse error below 3.3×10^{-13} and maximum complex error below 6.0×10^{-13} across six eligible lines. These are floating-point checks of an already proved identity.

5.5 Degenerate and exceptional cases

The assumptions in Theorem 5.1 are substantive.

- If $b = 0$, the process is deterministic. A stochastic phase is unnecessary; a time-independent polar graph may or may not contain the deterministic trajectory, depending on a and the interval.
- If $Z_0 = 0$, the normalization in Eq. 2 is undefined. One may choose another nonzero reference point, but that is a different representation problem.
- If $a = 0$ and $b/Z_0 \notin \mathbb{R}$, the theorem remains valid with $\lambda = 0$.

- If the line passes through the origin, a local polar chart can exist on one ray before a crossing, but it is not a global, lossless representation of the full Gaussian support.
- If $a/b \notin \mathbb{R}$, a time-dependent radius can represent each nonsingular fixed-time support line. It uses (t, θ) rather than one time-independent phase and fails when the moving line crosses the origin.

Figure 5 illustrates the last exception. It is a useful fallback, but it does not satisfy the one-phase problem posed in Definition Definition 3.1.

6 Canonical secant and constant-radius charts

6.1 Arithmetic Brownian factor as one Euler phase

Let

$$dX_t = \mu dt + \sigma dW_t, \quad X_0 \in \mathbb{R}, \quad (17)$$

and set $Y_t = X_t - X_0$. For any scale $c > 0$, define the auxiliary complex state

$$\mathcal{Z}_t = Z_0 \left(1 + i \frac{Y_t}{c} \right). \quad (18)$$

With

$$\theta_t = \arctan(Y_t/c) \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right), \quad (19)$$

the exact pathwise representation is

$$\boxed{\mathcal{Z}_t = Z_0 \sec \theta_t e^{i\theta_t}, \quad X_t = X_0 + c \tan \theta_t.} \quad (20)$$

This is Eq. 14 with $\rho = 1/c$ and $\phi = \pi/2$. If it is desirable for the real Brownian factor to be the real rather than imaginary coordinate, one may instead use $\tilde{Z}_t = ic + Y_t$ and $\tilde{\theta}_t = -\arctan(Y_t/c)$. These embeddings differ by a rigid rotation and scale.

Equation Eq. 20 answers the state-representation question in the requested form: neither t nor W_t appears separately on its right-hand side. They remain present in the law of θ , as they must. The sigma-fields generated by the current states agree,

$$\sigma(\theta_t) = \sigma(X_t),$$

and the completed path filtrations agree because both transforms are continuous and invertible.

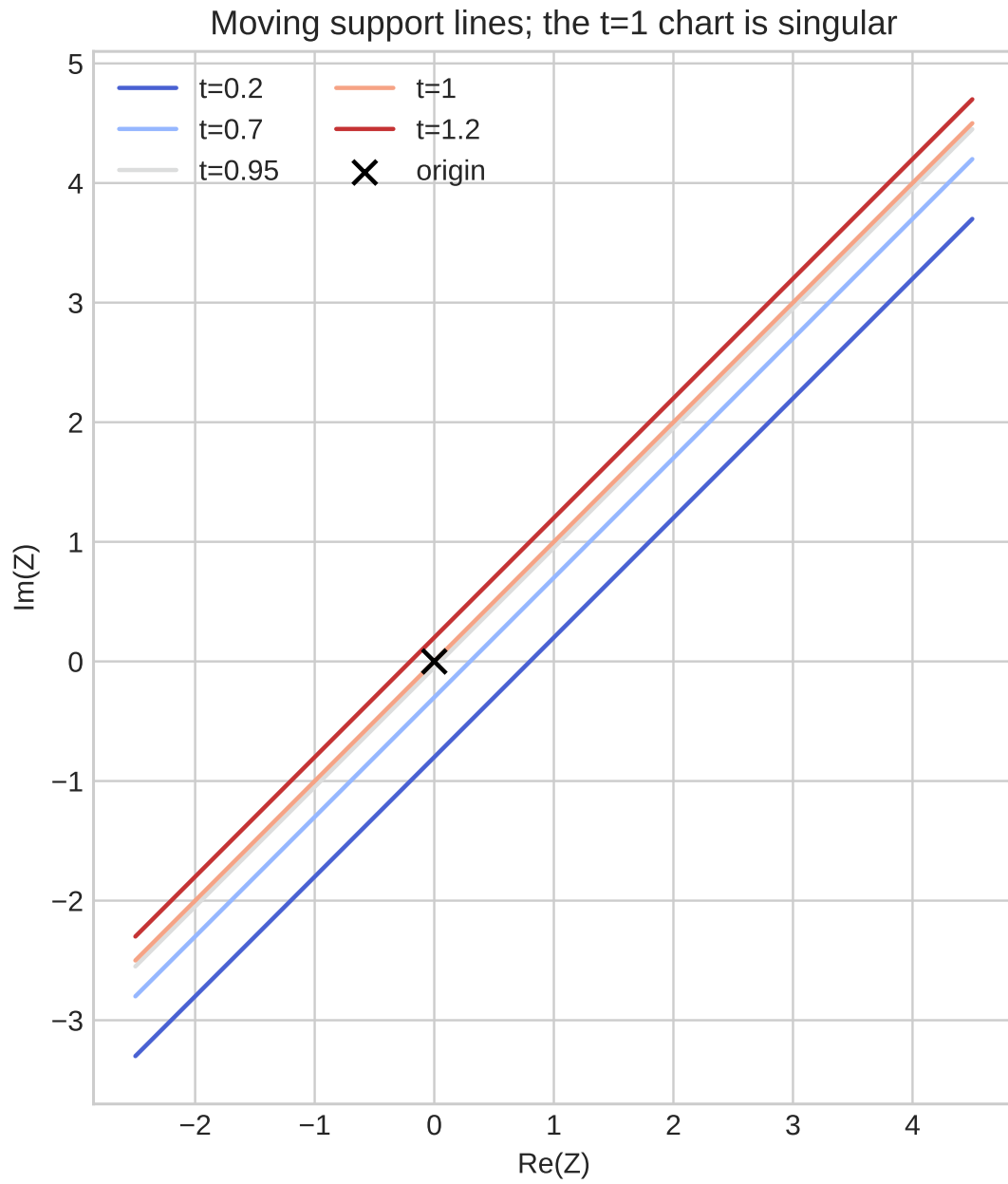


Figure 5: Time-dependent polar reconstruction for moving support lines and the singular origin-crossing time.

6.2 Geometry of the secant curve

Let $F(\theta) = \sec \theta e^{i\theta} = 1 + i \tan \theta$. Then

$$F'(\theta) = i \sec^2 \theta, \quad F''(\theta) = 2i \sec^2 \theta \tan \theta.$$

The signed curvature,

$$\kappa(\theta) = \frac{\operatorname{Im}\{\overline{F'(\theta)}F''(\theta)\}}{|F'(\theta)|^3},$$

is identically zero. Thus the algebraic factors “radius” and “rotation” conspire to trace a line. The stochastic phase is genuine—it is the angle of the point relative to the origin—but it is confined to one open half-turn.

The scale c does not change the information carried by the chart. It controls where physical values are allocated in phase:

$$\frac{d\theta}{dx} = \frac{c}{c^2 + x^2}.$$

Small c allocates more angular resolution near zero and compresses tails more aggressively; large c is closer to linear on a wider physical region. This tuning later affects PDE resolution and conditioning.

6.3 Cayley constant-radius alternative

A constant radius is possible after changing the represented state. Define

$$U_t = \frac{1 + iY_t/c}{1 - iY_t/c} = e^{i\Theta_t}, \quad \Theta_t = 2 \arctan(Y_t/c) \in (-\pi, \pi). \quad (21)$$

Because the denominator is the complex conjugate of the numerator, $|U_t| = 1$, and

$$Y_t = c \tan(\Theta_t/2).$$

The point $U = -1$ represents infinity and is inaccessible at every finite time. This construction keeps constant radius and losslessness on $S^1 \setminus \{-1\}$, but it represents a nonlinear transform of Eq. 18, not the original additive complex state.

The executed experiment in Figure 6 found maximum exact modulus defect 4.44×10^{-16} and inverse error 1.78×10^{-15} , whereas an unconstrained complex Euler control accumulated modulus error 0.252. That is a structure-preservation benefit for the transformed state. It does not make Θ standard Brownian motion on the circle: its angular diffusion is state dependent and vanishes near the missing point.

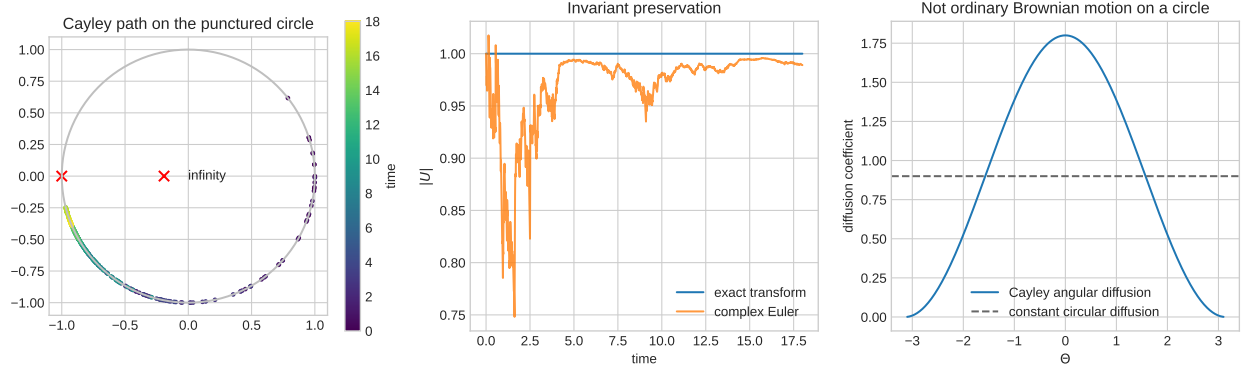


Figure 6: Cayley compactification, exact unit-modulus preservation, and the state-dependent angular diffusion near the missing point.

7 Phase diffusion, generator, kernel, and boundaries

7.1 Canonical Itô and Stratonovich equations

For $f(x) = \arctan(x/c)$,

$$f'(x) = \frac{c}{c^2 + x^2}, \quad f''(x) = -\frac{2cx}{(c^2 + x^2)^2}.$$

Applying Itô's formula to Eq. 17 and substituting $X - X_0 = c \tan \theta$ gives

$$d\theta_t = \left[\frac{\mu}{c} \cos^2 \theta_t - \frac{\sigma^2}{c^2} \sin \theta_t \cos^3 \theta_t \right] dt + \frac{\sigma}{c} \cos^2 \theta_t dW_t. \quad (22)$$

$$A(\theta) = \frac{\mu}{c} \cos^2 \theta - \frac{\sigma^2}{c^2} \sin \theta \cos^3 \theta, \quad B(\theta) = \frac{\sigma}{c} \cos^2 \theta. \quad (23)$$

Its Stratonovich form is simpler:

$$d\theta_t = \frac{\mu}{c} \cos^2 \theta_t dt + \frac{\sigma}{c} \cos^2 \theta_t \circ dW_t. \quad (24)$$

The Itô correction is not optional. Since $\frac{1}{2}BB' = -(\sigma^2/c^2) \sin \theta \cos^3 \theta$, converting Eq. 24 to Itô produces exactly the second term in Eq. 23.

Conversely, apply Itô's formula to $F(\theta) = 1 + i \tan \theta$ under Eq. 22. The nonlinear drift from $F' d\theta$ cancels the $\frac{1}{2}F'' d[\theta, \theta]$ term, leaving

$$d\{Z_0 F(\theta_t)\} = \frac{iZ_0}{c} (\mu dt + \sigma dW_t), \quad (25)$$

which reconstructs Eq. 18. This two-way calculation is the precise connection between Brownian quadratic variation and the complex curve: quadratic variation produces a geometric drift correction through F'' . It is not identified with i^2 .

7.2 General affine phase equation

For the eligible general line, define

$$g(\theta) := \frac{\rho \sin^2(\phi - \theta)}{\sin \phi}. \quad (26)$$

By Eq. 13 and Eq. 14, $d\theta/dx = g(\theta)$ and

$$g'(\theta) = -\frac{2\rho \sin(\phi - \theta) \cos(\phi - \theta)}{\sin \phi}.$$

Since $dx_t = \lambda dt + dW_t$,

$$\boxed{d\theta_t = \left[\lambda g(\theta_t) + \frac{1}{2} g'(\theta_t) g'(\theta_t) \right] dt + g(\theta_t) dW_t,} \quad (27)$$

or, in Stratonovich form,

$$d\theta_t = \lambda g(\theta_t) dt + g(\theta_t) \circ dW_t.$$

The sign of g records orientation. A scalar diffusion coefficient need not be declared positive; reversing W reverses its sign without changing the law.

7.3 Generator and phase-first specification

The canonical generator is

$$\boxed{\mathcal{L}_\theta h = A(\theta) h'(\theta) + \frac{1}{2} B(\theta)^2 h''(\theta),} \quad (28)$$

with A, B from Eq. 23. The martingale problem for Eq. 28 on $(-\pi/2, \pi/2)$ defines the phase law without naming t or W_t as state coordinates. Equivalently, its Markov transition kernel specifies the law. This is the rigorous interpretation of “ θ_t is the only stochastic state variable”: the process remains time-indexed and random, but its Markov state is one scalar.

7.4 Exact transition density and semigroup

Theorem 7.1 (Canonical transition kernel). *For $\sigma \neq 0$, $u, v \in (-\pi/2, \pi/2)$, and $h > 0$, the transition density of Eq. 22 is*

$$\boxed{p_h(v | u) = \frac{c \sec^2 v}{|\sigma| \sqrt{2\pi h}} \exp \left[-\frac{(c \tan v - c \tan u - \mu h)^2}{2\sigma^2 h} \right].} \quad (29)$$

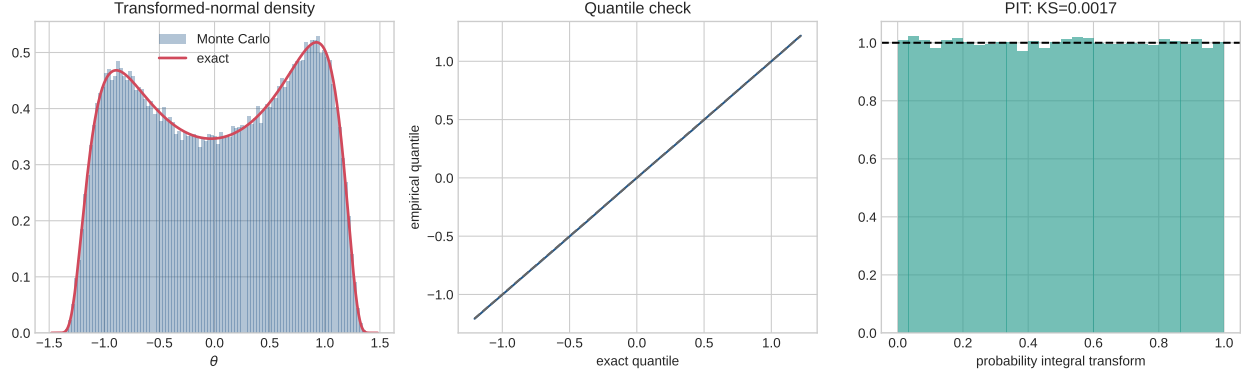


Figure 7: Transition-density normalization, probability-integral-transform diagnostics, and semigroup consistency.

Proof. Conditional on $\theta_s = u$,

$$X_{s+h} - X_0 = c \tan u + \mu h + \sigma \sqrt{h} \xi, \quad \xi \sim N(0, 1).$$

The inverse $v = \arctan((X - X_0)/c)$ has Jacobian $dX/dv = c \sec^2 v$, giving Eq. 29. $\square$

The exact step is therefore

$$\theta_{s+h} = \arctan \left(\tan \theta_s + \frac{\mu h}{c} + \frac{\sigma \sqrt{h}}{c} \xi \right). \quad (30)$$

For the general line, using the inverse $x(\theta)$ in Eq. 14,

$$p_h(v | u) = \frac{|x'(v)|}{\sqrt{2\pi h}} \exp \left[-\frac{(x(v) - x(u) - \lambda h)^2}{2h} \right], \quad (31)$$

where

$$|x'(v)| = \frac{|\sin \phi|}{\rho \sin^2(\phi - v)}.$$

Normalization follows by substituting $y = x(v)$. The Chapman–Kolmogorov identity follows from Gaussian convolution under the same substitution. Thus the “new” phase kernel is exactly the known Gaussian kernel pushed through a diffeomorphism; it is not a newly solved diffusion.

In Figure 7, numerical quadrature returned mass 0.9999999999999999, direct and composed semigroup values differed by 1.11×10^{-16} , and the probability-integral-transform KS statistic was 0.00173 with $p = 0.726$. A separate finite-difference forward-equation check had relative RMS residual 6.90×10^{-5} .

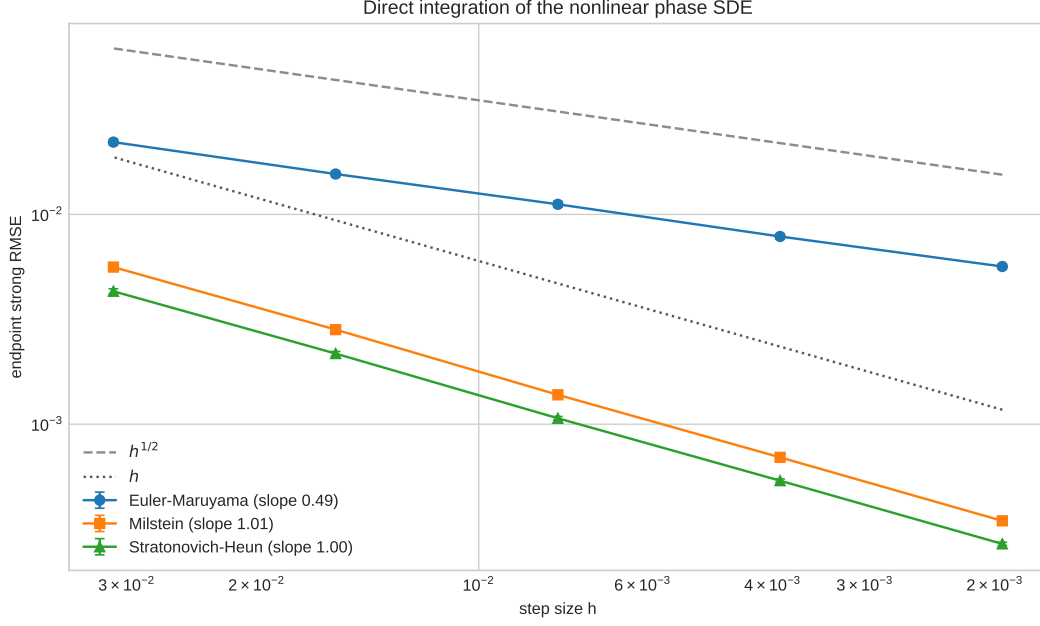


Figure 8: Strong convergence of independent nonlinear phase schemes and exact phase behavior near the natural boundaries.

7.5 Natural phase boundaries

The endpoints $\pm\pi/2$ correspond under $x = c \tan \theta$ to $\pm\infty$. For every finite T ,

$$\sup_{0 \leq t \leq T} |X_t| \leq |X_0| + |\mu|T + |\sigma| \sup_{0 \leq t \leq T} |W_t| < \infty \quad \text{a.s.}$$

Therefore neither phase endpoint is reached from the interior in finite time. Nor is either an entrance: for any compact $K \subset \mathbb{R}$ and $t > 0$,

$$\mathbb{P}_x(X_t \in K) \longrightarrow 0 \quad \text{as } x \rightarrow \pm\infty.$$

The Gaussian transition laws escape every compact set and have no probability limit on $\mathbb{R}$ from which a Feller entrance law can be defined. In the standard one-dimensional classification, both endpoints are natural (Feller 1954). Smooth monotone conjugacy preserves the classification, so

$$\boxed{-\frac{\pi}{2} \text{ and } \frac{\pi}{2} \text{ are inaccessible natural boundaries.}} \quad (32)$$

The same statement holds for the endpoints of I_ϕ . No reflecting or absorbing boundary condition should be imposed on the exact diffusion.

The numerical implications of Eq. 32 are visible in Figure 9. Exact transformed steps remain in the interval by construction. With $c = 0.25$, $\sigma = 1$, and 120,000 samples, unconstrained one-step Euler violation rates fell from 0.435 at $h = 0.25$ to 0.0271 at $h = 0.03125$. Bounded coordinates do not make a naive discretization automatically boundary preserving.

Zero observed rare endpoint events are not interpreted as zero probability. The notebook reports exact Gaussian tail probabilities and binomial upper confidence bounds together, as shown in Figure 10.

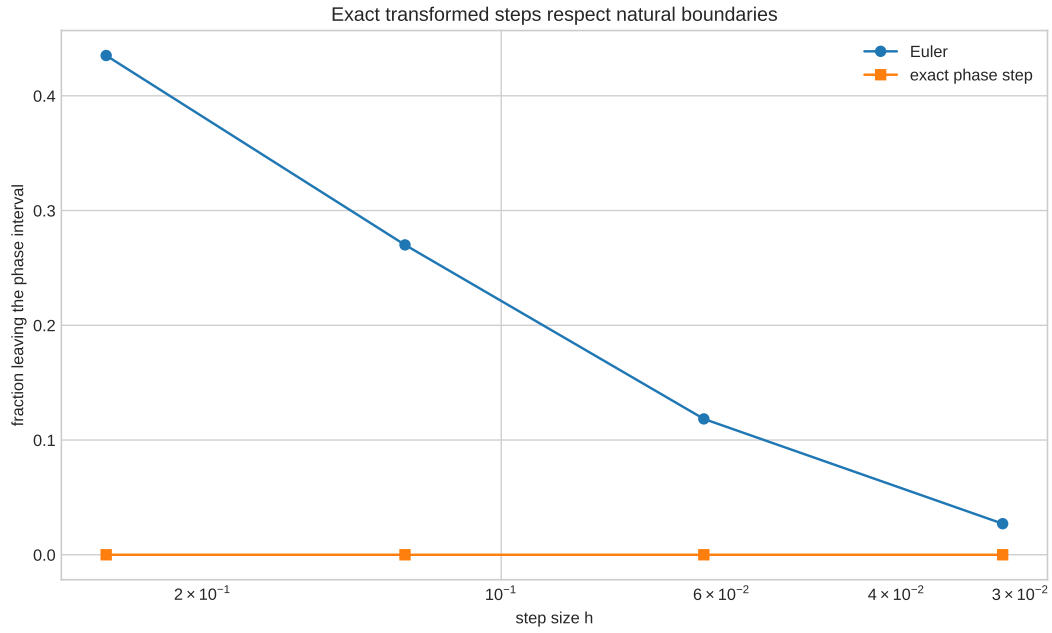


Figure 9: Boundary stress comparison between exact transformed steps and unconstrained Euler steps.

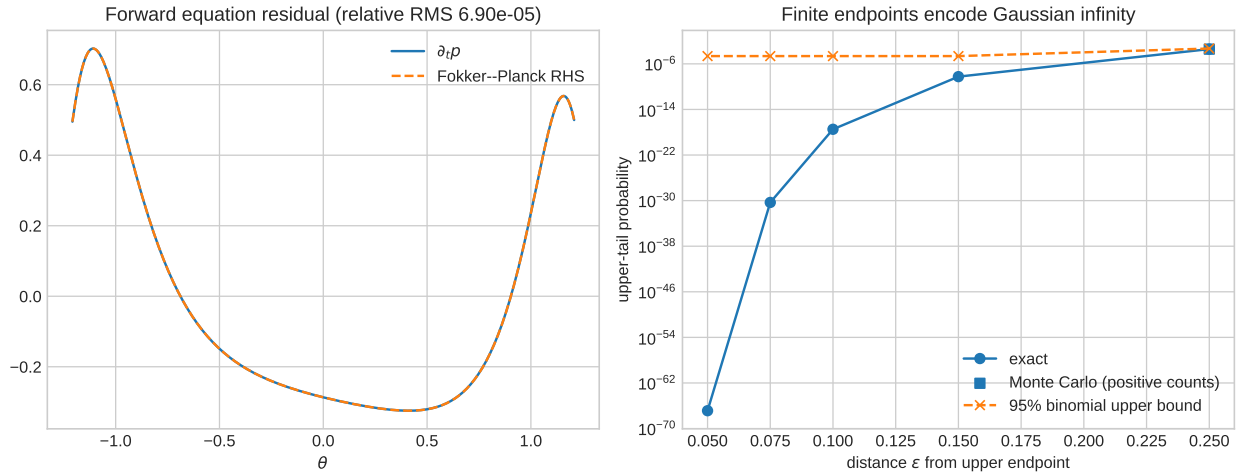


Figure 10: Small-time generator estimates, forward-equation residual, and endpoint-tail diagnostics.

8 Additive and multiplicative rigidity

8.1 Dynamics induced by a phase curve

Let

$$F(\theta) = r(\theta)e^{i\theta}, \quad r \in C^2(I; (0, \infty)),$$

and let

$$d\theta_t = m(\theta_t) dt + s(\theta_t) dW_t.$$

For $Z_t = Z_0 F(\theta_t)$, Itô's formula gives

$$\boxed{dZ_t = Z_0 \left(F' m + \frac{1}{2} F'' s^2 \right) (\theta_t) dt + Z_0 F'(\theta_t) s(\theta_t) dW_t.} \quad (33)$$

The diffusion coefficient already imposes a strong curve constraint; matching the drift supplies a second, independent compatibility equation.

Theorem 8.1 (Local additive rigidity). *Assume $F' \neq 0$ and $s \neq 0$. The curve-valued process in Eq. 33 satisfies*

$$dZ_t = a dt + b dW_t, \quad b \neq 0,$$

if and only if

$$F'(\theta)s(\theta) = \frac{b}{Z_0}, \quad \frac{a}{b} \in \mathbb{R}, \quad m(\theta) = \frac{a}{b} s(\theta) + \frac{1}{2} s(\theta)s'(\theta). \quad (34)$$

The image of F lies on a straight line. Every nonradial genuine polar member has local radius

$$r(\theta) = \frac{C}{\sin(\delta - \theta)}$$

on an interval where the expression is positive.

Proof. Diffusion matching yields $F' s = b/Z_0$. Differentiation gives $F'' = -(b/Z_0)s'/s^2$. Drift matching then reduces to

$$\frac{a}{b} = \frac{m}{s} - \frac{1}{2} s' \in \mathbb{R},$$

which is Eq. 34. Because s is real, F' always has the fixed complex direction b/Z_0 , so $F(I)$ is a line. Writing $F' = e^{i\theta}(r' + ir)$ and fixing the tangent argument δ gives $r'/r = \cot(\delta - \theta)$; integration yields the stated radius. $\square$

The global origin and losslessness restrictions in Theorem 5.1 remain necessary. Local tangent matching alone does not resolve them.

Theorem 8.2 (Local multiplicative rigidity). *Suppose instead that*

$$dZ_t = AZ_t dt + BZ_t dW_t, \quad B = \sigma + i\beta \neq 0.$$

A genuine phase with $s \neq 0$ exists if and only if

$$\boxed{\beta \neq 0, \quad s(\theta) \equiv \beta, \quad r(\theta) = r_0 e^{k\theta}, \quad k = \frac{\sigma}{\beta},} \quad (35)$$

and $m(\theta) \equiv m_0 \in \mathbb{R}$ satisfies

$$\boxed{A = (k + i)m_0 + \frac{1}{2}(k + i)^2 \beta^2.} \quad (36)$$

Equivalently,

$$A - \frac{1}{2}B^2 \in B\mathbb{R}. \quad (37)$$

Proof. Diffusion matching gives $(F'/F)s = B$. Since

$$\frac{F'}{F} = \frac{r'}{r} + i,$$

imaginary parts force $s = \beta \neq 0$, and real parts give $r'/r = \sigma/\beta = k$. Thus $r = r_0 e^{k\theta}$. Substituting $F'/F = k + i$ and $F''/F = (k + i)^2$ into the drift equation forces constant m_0 and Eq. 36. The equivalent logarithmic condition follows from $B = (k + i)\beta$. $\square$

The line and spiral dictionary is therefore:

Table 2: Additive and multiplicative rigidity.

Constant-coefficient law	Forced curve	Drift compatibility
$dZ = a dt + b dW$	Straight affine line; secant-type polar radius	$a/b \in \mathbb{R}$ and Eq. 34
$dZ/Z = A dt + B dW$	Logarithmic spiral $r = r_0 e^{(\text{Re } B/\text{Im } B)\theta}$	$A - \frac{1}{2}B^2 \in B\mathbb{R}$

The numerical residuals in Figure 11 are controls: 4.90×10^{-17} for the line/additive match and 1.09×10^{-16} for the spiral/multiplicative match. They illustrate, but do not prove, the coefficient-matching theorems.

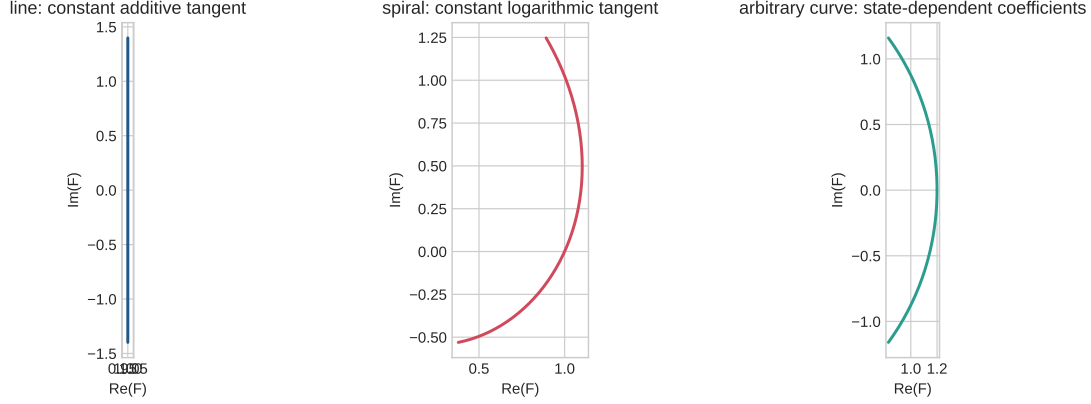


Figure 11: Residual checks distinguishing additive lines, multiplicative logarithmic spirals, and a generic polar curve.

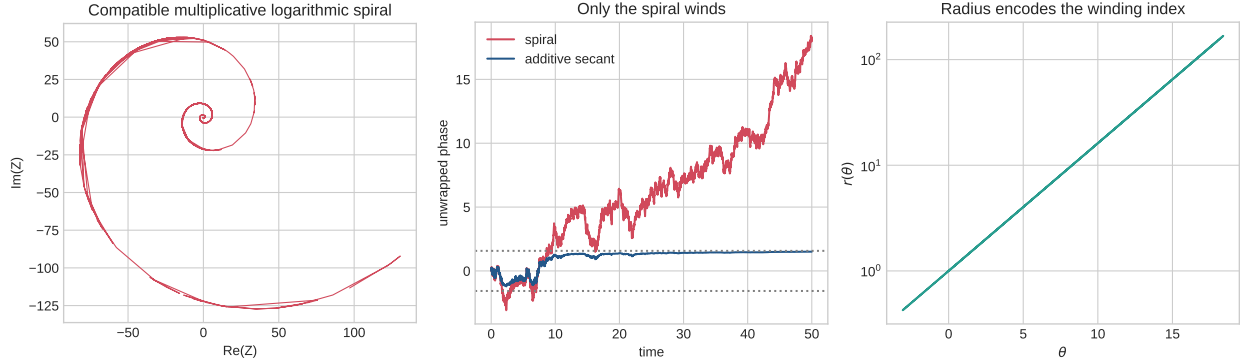


Figure 12: Additive half-turn geometry versus compatible multiplicative logarithmic-spiral winding.

8.2 Winding requires a multiplicative model

For

$$Z_t = Z_0 \exp(\kappa t + BW_t), \quad B = \sigma + i\beta, \quad (38)$$

condition Eq. 37 says $\kappa = \gamma B$ for some $\gamma \in \mathbb{R}$. When $\beta \neq 0$,

$$\theta_t = \beta(\gamma t + W_t), \quad \boxed{Z_t = Z_0 \exp\left(\frac{\sigma}{\beta}\theta_t\right) e^{i\theta_t}.} \quad (39)$$

Unlike the additive secant phase, θ ranges over the real line and can wind indefinitely. The price is a change in the stochastic model from additive affine dynamics to a compatible multiplicative exponential.

The simulated spiral in Figure 12 spanned 21.49 radians (3.42 turns), while the additive phase span remained $2.72 < \pi$. This is a model distinction, not evidence that one representation is a rewriting of the other.

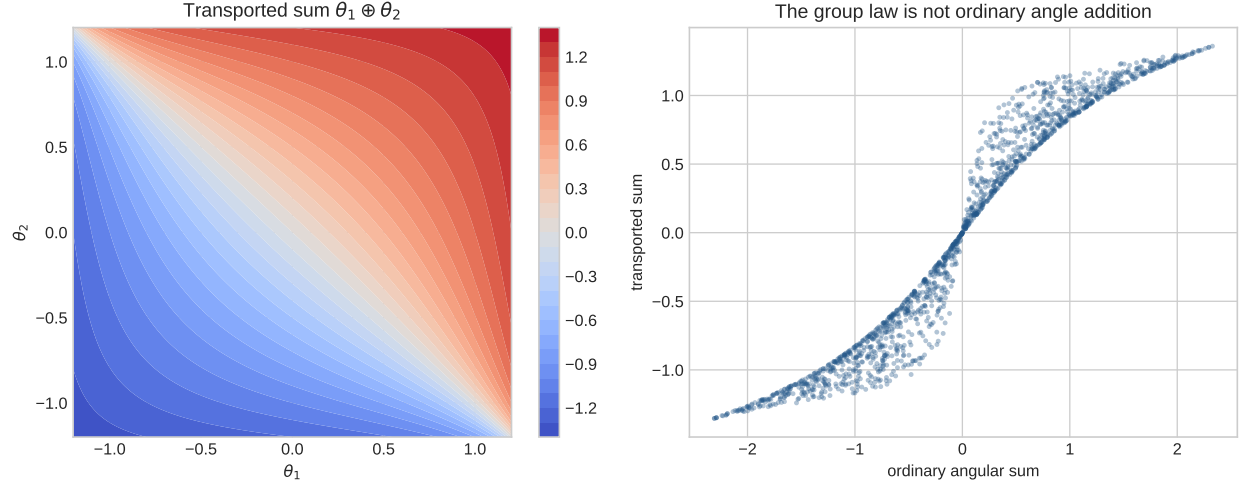


Figure 13: The transported secant and Cayley group laws compared with ordinary angular addition.

8.3 Transported state group

The canonical chart

$$\Phi_c(x) = \arctan(x/c)$$

transports real addition to

$$\boxed{\theta_1 \oplus \theta_2 = \arctan(\tan \theta_1 + \tan \theta_2).} \quad (40)$$

Thus

$$((-\pi/2, \pi/2), \oplus) \cong (\mathbb{R}, +).$$

The operation is associative and commutative, has identity zero and inverse $-\theta$, and exactly combines scalar increments inside the bounded chart. It is not ordinary angular addition. Indeed,

$$(1 + ix/c)(1 + iy/c) = 1 + i(x + y)/c - xy/c^2,$$

so the secant embedding does not turn addition into ordinary complex multiplication.

The notebook verified the tangent-addition identity to 3.11×10^{-15} and associativity to 1.67×10^{-15} , while the RMS discrepancy from ordinary angular addition was 0.288. The group structure comes from conjugated real addition.

There is no continuous injective homomorphism from $(\mathbb{R}, +)$ into $(S^1, \cdot)$. Every continuous one-parameter subgroup of S^1 is $x \mapsto e^{i\kappa x}$; it is constant if $\kappa = 0$ and periodic otherwise. Consequently, constant radius, lossless recovery of an unrestricted real state, and ordinary complex multiplication cannot all hold. The Cayley chart retains constant radius and losslessness but uses a transported group law. The logarithmic spiral retains losslessness and ordinary multiplicative composition by allowing the radius to vary.

9 Risk-neutral pricing conjugacy

9.1 Normal factors and the auxiliary complex coordinate

Consider a real forward F under its T -forward measure $\mathbb{Q}^T$:

$$dF_t = \sigma_N dW_t^{\mathbb{Q}^T}, \quad 0 \leq t \leq T, \quad (41)$$

where $\sigma_N \geq 0$ is constant for the basic Bachelier model. Let $D(t, T)$ be the deterministic discount factor. The normal model permits negative levels, which is a limitation for some assets and an intended feature in other markets (Choi et al. 2022).

For a chosen coordinate scale $c > 0$, set

$$\theta_t = \arctan(F_t/c), \quad \mathcal{Z}_t = 1 + iF_t/c = \sec \theta_t e^{i\theta_t}. \quad (42)$$

The factor F remains the priced real state. The complex $\mathcal{Z}$ is neither a numeraire nor a tradeable security. It is a lossless chart, and

$$F_t = c \tan \theta_t.$$

If the physical-measure factor has drift and an equivalent Girsanov change removes or changes that drift, the diffusion direction is unchanged (Girsanov 1960). The canonical chart remains eligible. More generally, an affine complex model satisfying Eq. 11 retains its fixed diffusion line under equivalent changes whose drift remains in the same real span.

9.2 Conditional-expectation invariance

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be Borel and integrable under the conditional law, and let the short rate $r : [0, T] \rightarrow \mathbb{R}$ be deterministic (or work in numeraire units, where the discount term is absent). For a real Markov factor

$$dX_t = \mu_{\mathbb{Q}} dt + \sigma dW_t^{\mathbb{Q}},$$

define

$$V(t, x) = \mathbb{E}^{\mathbb{Q}} \left[\exp \left(- \int_t^T r(s) ds \right) g(X_T) \mid X_t = x \right]. \quad (43)$$

Set

$$v(t, \theta) = V(t, c \tan \theta). \quad (44)$$

Because $x \leftrightarrow \theta$ is deterministic and bijective,

$$\boxed{V(t, X_t) = v(t, \theta_t)} \quad (45)$$

pathwise. This equality is stronger than equality in distribution: every state and every payoff sample is the same after exact inversion.

Theorem 9.1 (Generator and pricing-PDE conjugacy). *Suppose $V \in C^{1,2}$ solves*

$$V_t + \mu_{\mathbb{Q}} V_x + \frac{1}{2} \sigma^2 V_{xx} - r(t)V = 0, \quad V(T, x) = g(x). \quad (46)$$

Then v in Eq. 44 solves

$$\boxed{v_t + A(\theta)v_{\theta} + \frac{1}{2}B(\theta)^2v_{\theta\theta} - r(t)v = 0,} \quad (47)$$

with

$$A(\theta) = \frac{\mu_{\mathbb{Q}}}{c} \cos^2 \theta - \frac{\sigma^2}{c^2} \sin \theta \cos^3 \theta, \quad B(\theta) = \frac{\sigma}{c} \cos^2 \theta,$$

and terminal condition

$$v(T, \theta) = g(c \tan \theta).$$

Proof. One route applies Feynman–Kac/Kac reasoning to the phase generator Eq. 28. Directly, from $x = c \tan \theta$,

$$v_{\theta} = V_x c \sec^2 \theta, \quad (48)$$

and

$$v_{\theta\theta} = V_{xx} c^2 \sec^4 \theta + 2V_x c \tan \theta \sec^2 \theta. \quad (49)$$

Substitution of Eq. 48 and Eq. 49 into $Av_{\theta} + \frac{1}{2}B^2v_{\theta\theta}$ cancels the two first-derivative Itô terms and gives

$$Av_{\theta} + \frac{1}{2}B^2v_{\theta\theta} = \mu_{\mathbb{Q}}V_x + \frac{1}{2}\sigma^2V_{xx}. \quad (50)$$

Equation Eq. 47 follows from Eq. 46. $\square$

This is operator conjugacy. It preserves the spectrum and semigroup in the appropriate transformed function spaces, but a particular discretized matrix can have different resolution and conditioning.

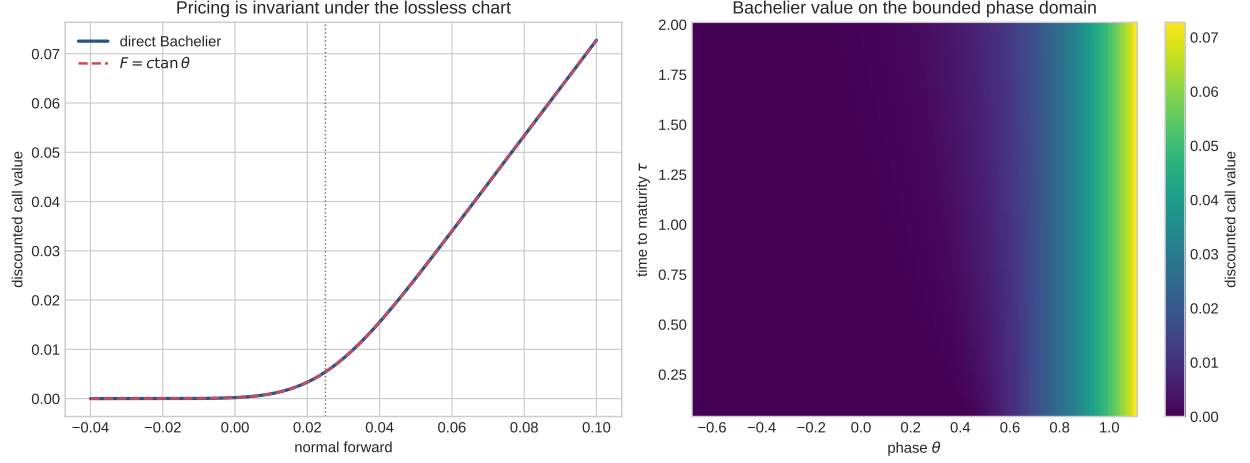


Figure 14: Bachelier call prices in the scalar and phase coordinates, and the value surface over phase and maturity.

9.3 Bachelier European calls

For time to maturity $\tau = T - t$, normal volatility σ_N , forward F , and strike K , let

$$s = \sigma_N \sqrt{\tau}, \quad d = \frac{F - K}{s}.$$

For $s > 0$, the discounted Bachelier call is

$$C_B(t, F) = D(t, T) [(F - K)\Phi(d) + s\varphi(d)]. \quad (51)$$

At $s = 0$, it is $D(t, T)(F - K)^+$. In phase coordinates,

$$c_B(t, \theta) = D(t, T) [(c \tan \theta - K)\Phi(d_\theta) + s\varphi(d_\theta)], \quad (52)$$

where

$$d_\theta = \frac{c \tan \theta - K}{s}.$$

Equation Eq. 52 is a composition of the established Bachelier formula, not a new option-pricing formula.

The benchmark behind Figure 14 used $c = 0.05$, $K = 0.025$, $\sigma_N = 0.0125$, $\tau = 1.25$, and $D = 0.97$. Across 401 forward values from -0.04 to 0.10 , the maximum absolute difference between Eq. 51 and Eq. 52 was 2.78×10^{-17} . The calculation includes negative forwards and both in- and out-of-the-money regimes.

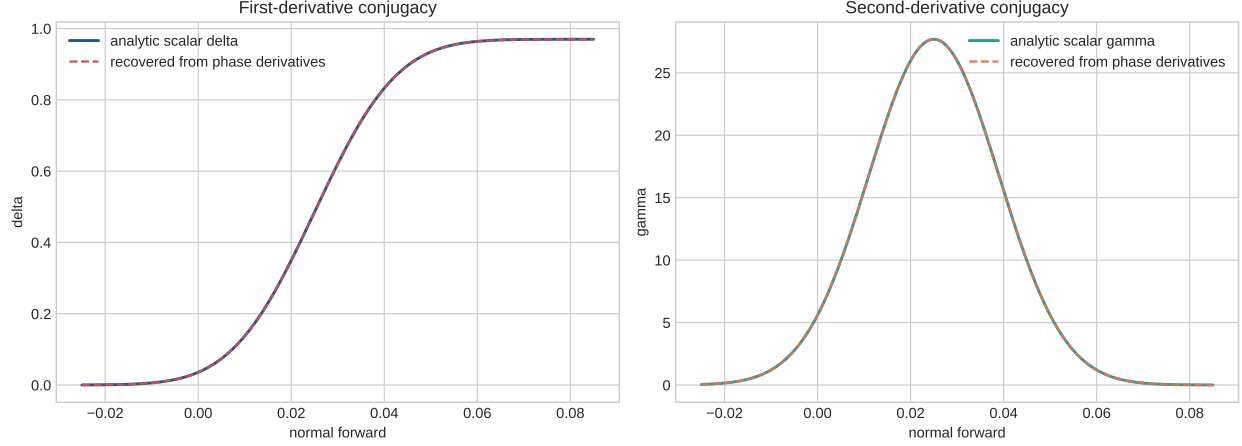


Figure 15: Bachelier delta and gamma recovered from finite-difference phase derivatives.

9.4 Delta, gamma, and hedge conversion

For the scalar normal call,

$$\Delta_F = \frac{\partial C_B}{\partial F} = D(t, T)\Phi(d), \quad \Gamma_F = \frac{\partial^2 C_B}{\partial F^2} = D(t, T)\frac{\varphi(d)}{s}. \quad (53)$$

For a general value $v(t, \theta) = V(t, c \tan \theta)$, inverse chain rules give

$$V_x = \frac{\cos^2 \theta}{c} v_\theta, \quad (54)$$

and

$$V_{xx} = \frac{\cos^4 \theta v_{\theta\theta} - 2 \sin \theta \cos^3 \theta v_\theta}{c^2}. \quad (55)$$

The second term in Eq. 55 is a Hessian correction. Omitting it is the deterministic analogue of omitting the Itô correction in a transformed generator.

In Figure 15, phase derivatives were computed by centered differences with step 7.5×10^{-5} and converted by Eq. 54 and Eq. 55. Against independent analytic Greeks, the maximum absolute errors were 1.76×10^{-8} for delta and 5.81×10^{-7} for gamma.

A self-financing hedge is coordinate invariant when the Jacobian is handled correctly. If the hedge ratio is V_x , using v_θ directly would be wrong by the factor $d\theta/dx = \cos^2 \theta/c$. The auxiliary complex coordinate does not create another hedge instrument or complete an otherwise incomplete market.

9.5 Pricing-PDE verification

Under the driftless normal forward model and with discounting factored out, the forward-maturity equation is

$$V_\tau = \frac{1}{2} \sigma_N^2 V_{FF}. \quad (56)$$

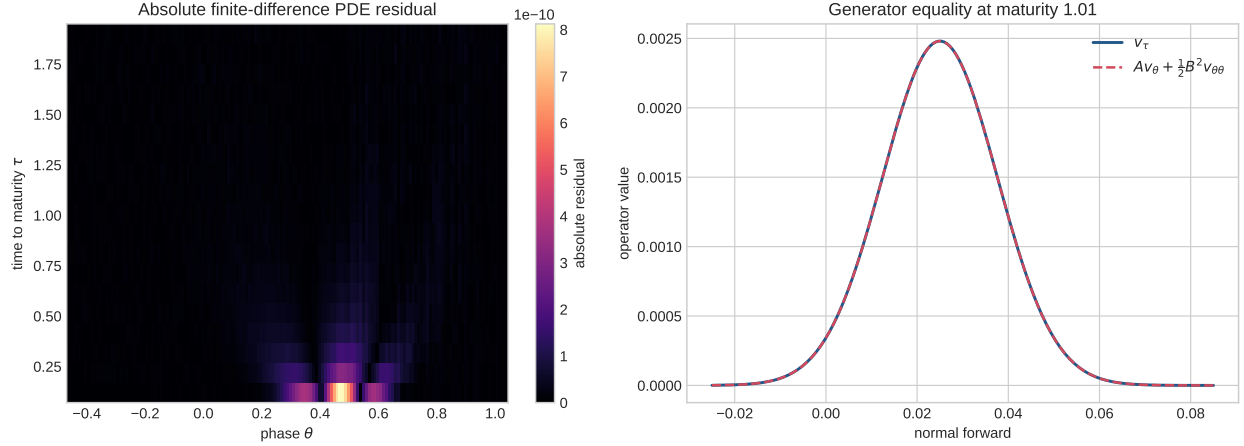


Figure 16: Finite-difference verification of Bachelier pricing-PDE conjugacy.

The phase equation is

$$v_\tau = A(\theta)v_\theta + \frac{1}{2}B(\theta)^2v_{\theta\theta}, \quad (57)$$

where

$$A(\theta) = -\frac{\sigma_N^2}{c^2} \sin \theta \cos^3 \theta, \quad B(\theta) = \frac{\sigma_N}{c} \cos^2 \theta.$$

Independent centered differences in phase and maturity were evaluated at 2,869 grid points in Figure 16. The maximum absolute residual was 8.12×10^{-10} and the RMS residual was 5.51×10^{-11} . The maximum relative residual was about 2.05×10^{-3} only after dividing by operator values as small as 10^{-8} ; absolute residuals are the stable primary metric.

9.6 Payoff regularity and compactified boundaries

The coordinate change does not smooth a payoff kink. A call kink at $x = K$ is moved to

$$\theta_K = \arctan(K/c).$$

For payoffs with polynomial growth, the transformed terminal condition may diverge as $\theta \rightarrow \pm\pi/2$ because $c \tan \theta \rightarrow \pm\infty$. Natural boundaries remove the need for a probabilistic reflecting or absorbing condition, but a numerical method on a truncated phase grid still needs consistent asymptotic treatment. The concentration of grid points and the degeneracy $B(\theta) \rightarrow 0$ can help or hurt depending on the payoff, basis, scale, and solver.

10 Numerical methodology and evidence

10.1 Reproducible design

The computational record is an executed Jupyter notebook generated by a plain Python builder. It imports a tested mathematical module rather than redefining the core transforms in disconnected cells. The environment

records Python 3.12, NumPy 2.5.1, SciPy 1.18.0, pandas 3.0.3, Matplotlib 3.11.1, and nbformat 5.10.4. The primary and replication seed families are 20260725 and 20360725.

Comparisons use shared random streams when samplewise equality or strong error is under study. Monte Carlo standard errors are retained. Every publication figure is saved as vector PDF and raster PNG, and every numerical assertion is saved as CSV or JSON. The notebook terminates unless all 22 figures exist in both formats and its accuracy gates pass.

The experiments are organized by evidentiary role:

- deterministic grids check exact reconstruction and inverse formulas;
- independently advanced nonlinear schemes check phase SDE coefficients;
- quadrature, simulation, and finite differences check the kernel, generator, semigroup, forward equation, and pricing PDE;
- matched scalar/phase streams test estimator invariance; and
- negative controls test dimension, path dependence, complex derivatives, winding, and numerical conditioning.

Simulation supports and stress-tests proofs; it does not establish a theorem or a novelty claim.

10.2 Nonlinear SDE convergence

Euler–Maruyama, Milstein, and Stratonovich–Heun were advanced directly from Eq. 22 or Eq. 24. The exact arctangent update was used only as a shared-path endpoint reference. With $\mu = 0.3$, $\sigma = 0.8$, $c = 1.25$, $T = 1$, 4,000 paths, and step counts 32, ..., 512, the fitted strong slopes were:

Table 3: Shared-path strong-convergence estimates.

Method	Base-seed slope	Replication slope	Base RMSE at 512 steps	Replication RMSE
Euler–Maruyama	0.493	0.510	5.65×10^{-3}	5.45×10^{-3}
Milstein	1.006	1.011	3.46×10^{-4}	3.51×10^{-4}
Stratonovich–Heun	1.001	1.000	2.69×10^{-4}	2.71×10^{-4}

The expected orders 1/2 and 1 appear in two independent seed families, as visualized in Figure 8. This is a non-circular check of the derived coefficients. The exact step remains merely the known Gaussian step conjugated by a known inverse; it is not a general exact-simulation algorithm in the sense of (Beskos and Roberts 2005).

10.3 Heat-equation compactification benchmark

To isolate numerical effects from financial payoff singularities, the notebook also solved

$$\partial_t u = \frac{1}{2} \partial_{xx} u, \quad u(0, x) = e^{-x^2}, \quad (58)$$

at $T = 0.35$, with physical truncation $[-8, 8]$. Direct- x and $x = c \tan \theta$ grids used matched total point counts, and errors were compared on the common region $|x| \leq 4$.

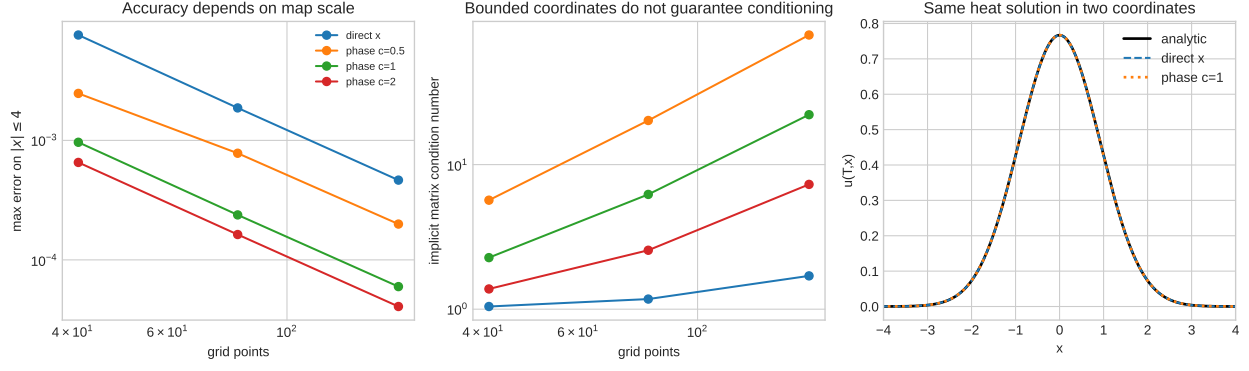


Figure 17: Accuracy and conditioning trade-offs for direct and arctangent-compactified heat-equation grids.

Table 4: Matched-grid compactification benchmark on $|x| \leq 4$.

Coordinate	c	Points	Maximum error	RMS error	Condition proxy
direct x	—	161	4.65×10^{-4}	1.78×10^{-4}	1.70
phase	0.5	161	1.99×10^{-4}	6.04×10^{-5}	78.8
phase	1.0	161	5.98×10^{-5}	2.28×10^{-5}	22.1
phase	2.0	161	4.07×10^{-5}	2.55×10^{-5}	7.30

At 161 points, the best mapped maximum error in Table 4 was about 11.4 times smaller than the direct-grid error, but even the best-conditioned mapped case had a condition proxy about 4.3 times larger. At $c = 0.5$, conditioning deteriorated far more. The qualified conclusion agrees with the mapped-domain literature (Grosch and Orszag 1977; Boyd 1987): compactification can reallocate resolution, but its value is scale-, solution-, discretization-, and metric-dependent.

10.4 Monte Carlo and quasi-Monte Carlo invariance

For any terminal payoff g ,

$$g(X_T) = g(c \tan \theta_T) \quad (59)$$

sample by sample. Consequently, if the same random stream is used,

$$\widehat{V}_X = \widehat{V}_\theta, \quad \widehat{\text{Var}}_X = \widehat{\text{Var}}_\theta. \quad (60)$$

In Figure 18, the direct and phase Bachelier-payoff estimators both returned 0.363588299584, sample variance 0.320662659809, and standard error 0.00110599773. State and payoff discrepancies were at most 1.78×10^{-15} . Scrambled Sobol estimates and an importance-sampled rare-event estimate also agreed. The phase route retains one random dimension and adds an arctangent/tangent round trip. Variance reduction could arise only from another algorithmic change—such as a new proposal, control variate, smoothing, stratification, or basis—not from exact relabeling alone.

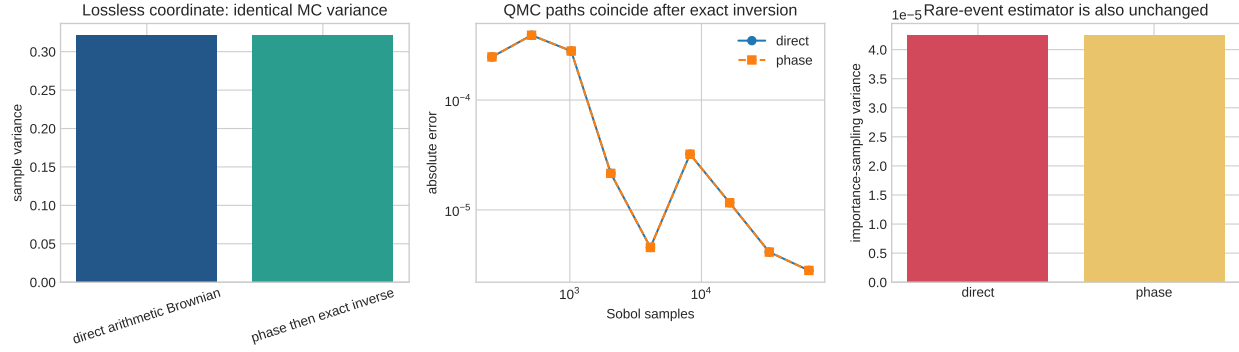


Figure 18: Ordinary Monte Carlo, scrambled Sobol, and importance-sampling comparisons under the exact phase round trip.

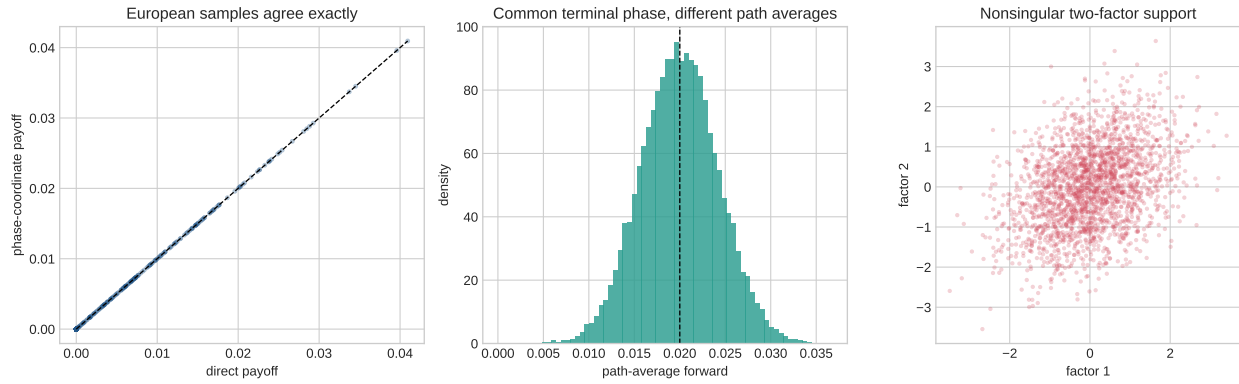


Figure 19: Financial scope controls for European payoff invariance, pinned-path dependence, and two-factor support.

10.5 Path dependence and multifactor controls

A terminal phase is sufficient for a terminal one-factor payoff because it determines X_T . It is not sufficient for a generic path functional. To make the distinction exact, the notebook simulated 28,000 Brownian bridges pinned at the same terminal forward. Their terminal phase spread was 2.78×10^{-16} , yet their time-average forward had standard deviation 0.00427 and 1%–99% quantiles 0.0101 and 0.0300.

A second control simulated a nonsingular correlated two-factor Gaussian law. Its smallest covariance eigenvalue was 0.654, its determinant was 0.879, and its rank was two. No smooth one-dimensional phase chart can be a diffeomorphism onto an open two-dimensional support.

Figure 19 separates three claims:

Table 5: Financial scope controls.

Claim	Result
Exact phase changes a European payoff sample	Refuted; maximum difference 1.39×10^{-17}
Terminal phase determines a path average	Refuted by pinned bridges
One smooth phase losslessly represents a nonsingular two-factor law	Refuted by covariance rank two

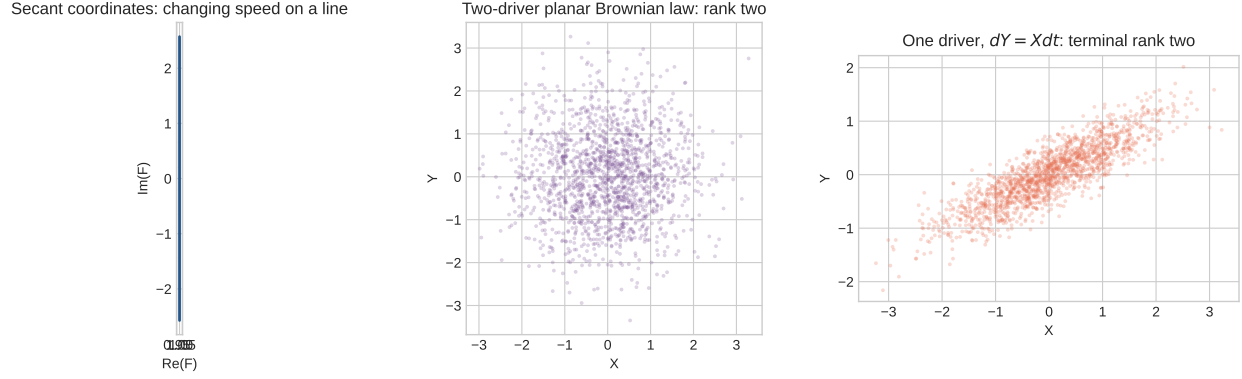


Figure 20: Curve tangents and normals, instantaneous rank-one covariance, and higher-dimensional negative controls.

The full phase *path* is equivalent to the full scalar path. The negative result concerns replacing path history by the terminal phase alone.

11 Stochastic dimension and vector geometry

11.1 Tangent, normal, and instantaneous covariance

For a C^2 phase curve $F : I \rightarrow \mathbb{C}$, the complex derivative $F'(\theta)$ is a tangent vector under $\mathbb{C} \simeq \mathbb{R}^2$, and $iF'(\theta)$ is its quarter-turned normal. If

$$d\theta_t = m(\theta_t)dt + s(\theta_t)dW_t,$$

then the real Cartesian diffusion vector of $Z_t = Z_0 F(\theta_t)$ is

$$q(\theta) = s(\theta) \begin{pmatrix} \operatorname{Re}\{Z_0 F'(\theta)\} \\ \operatorname{Im}\{Z_0 F'(\theta)\} \end{pmatrix}.$$

Its instantaneous covariance density is the outer product

$$\boxed{\Sigma(\theta) = q(\theta)q(\theta)^\top, \quad \operatorname{rank} \Sigma(\theta) \leq 1.} \tag{61}$$

Multiplication by i rotates q but does not supply an independent increment. To obtain independent tangential and normal Brownian forcing, the model needs another driver or an explicitly higher-rank diffusion matrix.

11.2 Four notions of dimension

The phrase “one-dimensional noise” can refer to different objects. It is important to distinguish:

1. the number of Brownian drivers;
2. the rank of the instantaneous diffusion covariance;
3. the dimension of the fixed-time support; and
4. the dimension of a sufficient Markov state.

For the affine process classified here, all four relevant state notions are one-dimensional. This is not a universal implication of having one driver. For example,

$$dX_t = dW_t, \quad dY_t = X_t dt \quad (62)$$

has one Brownian driver and instantaneous covariance rank one, but

$$\text{Cov}(X_1, Y_1) = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{pmatrix}, \quad \det = \frac{1}{12} > 0. \quad (63)$$

The drift propagates noise into a second state direction. This elementary hypoelliptic example reflects the broader support and bracket phenomena studied by Stroock–Varadhan and Hörmander (Stroock and Varadhan 1972; Hörmander 1967). The notebook’s empirical covariance was within 8.55×10^{-3} of Eq. 63 and had rank two, as did a planar Brownian control.

Thus neither shortcut is valid:

one driver $\nRightarrow$ one-dimensional terminal support,

and

complex notation $\nRightarrow$ two stochastic dimensions.

11.3 What vector geometry does contribute

The complex chart provides concise tangent-normal bookkeeping. In more general curve-valued models it can organize projections, curvature drift, constraint error, and invariant-preserving updates. In the present additive case, however, the secant curve has zero curvature and all tangents are parallel. The geometry explains the model’s rank-one structure; it does not enlarge it.

The rigidity theorems convert coefficient matching into geometry:

- constant additive noise requires constant tangent direction, hence a line;
- constant multiplicative noise requires constant logarithmic tangent F'/F , hence a logarithmic spiral; and
- a general polar curve induces state-dependent complex coefficients.

This is a useful coordinate interface to existing vector stochastic calculus, not a new vector calculus.

12 Complex differentiation and analytic implications

12.1 A curve determines only tangential derivatives

A complex derivative at an interior point of an open subset of $\mathbb{C}$ requires compatible limits from all directions. Data restricted to the one-dimensional state curve determine only a tangential derivative.

On the canonical line $\{1 + iy : y \in \mathbb{R}\}$, consider

$$H_0(z) = 0, \quad H_1(z) = \text{Re } z - 1.$$

Both functions vanish on the entire state line, yet their normal derivatives are 0 and 1. Hence the stochastic state data do not select a unique off-curve extension, much less a unique holomorphic extension.

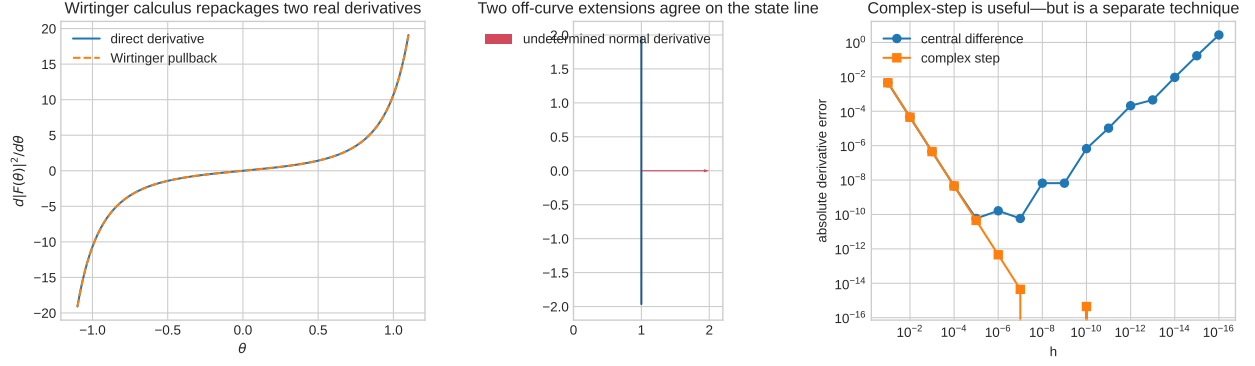


Figure 21: Off-curve derivative nonuniqueness, Wirtinger pullback, and the separate complex-step benchmark.

12.2 Wirtinger calculus

For a real-differentiable $H(z, \bar{z})$, define

$$H_z = \frac{1}{2}(H_x - iH_y), \quad H_{\bar{z}} = \frac{1}{2}(H_x + iH_y).$$

Along $z = F(\theta)$,

$$\frac{d}{d\theta} H(F(\theta), \overline{F(\theta)}) = H_z F'(\theta) + H_{\bar{z}} \overline{F'(\theta)}. \quad (64)$$

This compactly packages two real derivatives (Brandwood 1983). It becomes holomorphic calculus only when $H_{\bar{z}} = 0$ on an open domain. The notebook checked Eq. 64 for $H(z, \bar{z}) = |z|^2$ to 3.55×10^{-15} , while the two extensions above agreed exactly on the curve and retained different normal derivatives.

12.3 Complex-step differentiation and transform methods

For a real function with a suitable analytic extension,

$$f'(x) \approx \frac{\text{Im } f(x + ih)}{h}$$

can avoid subtractive cancellation (Squire and Trapp 1998). That advantage comes from the analytic extension and the complex-step algorithm. The phase chart does not supply off-curve values or guarantee analyticity of a payoff.

Similarly, complex methods in models such as Heston and in Fourier pricing derive their power from characteristic functions, contour choices, damping, and transform inversion (Heston 1993; Carr and Madan 1999). Writing a real Bachelier state as $\sec \theta e^{i\theta}$ neither replaces nor automatically improves those methods.

Figure 21 summarizes the boundary: complex notation is valuable for organization and for methods whose own analytic assumptions hold; Euler form alone creates no holomorphic derivative.

13 Computational implications and research agenda

13.1 What is analytically gained

The phase chart supplies exact formulas for an already solvable affine diffusion:

- a bounded state coordinate and explicit inverse;
- an autonomous nonlinear SDE;
- an exact transition density and update;
- natural-boundary behavior;
- a conjugate generator and pricing PDE;
- exact transformed Bachelier values and Greeks; and
- group and rigidity interpretations.

This package can simplify reasoning when a bounded chart, polar geometry, or state-dependent grid is desirable. It also creates a demanding but exactly solvable benchmark for nonlinear SDE schemes.

It does **not** turn a previously unsolved stochastic problem into a closed form. Every canonical analytic formula is inherited from arithmetic Brownian motion through a diffeomorphism. An invertible coordinate change preserves information and exact option values. Any computational improvement must enter through the interaction between the coordinate and an additional numerical choice.

13.2 Qualified opportunities

13.2.1 Adaptive and spectral PDE methods

The map scale c provides a principled way to concentrate resolution near a region of financial interest while retaining an unbounded physical domain. Rational spectral bases induced by related maps have established theory (Boyd 1987). A useful next result would optimize c jointly with payoff regularity, tail asymptotics, grid order, and a conditioning constraint, rather than report a single favorable error.

13.2.2 Structure-preserving phase integration

Exact transformed steps preserve the natural interval, and exact Cayley steps preserve unit modulus. For models without an exact inverse, geometric and symmetry-adapted schemes may exploit the same invariants (Malham and Wiese 2008; De Vecchi et al. 2019). A rigorous extension would compare weak and strong error at matched computational cost while measuring constraint defect separately.

13.2.3 Financial normal-model calculations

The phase can be useful when normal factors range over wide positive and negative values and a bounded PDE coordinate is operationally convenient. It does not change calibration targets or risk measures. Empirical work would need to compare stability of calibration, extrapolation, and Greeks under specified market data, boundary conventions, and regularization.

13.2.4 Higher-dimensional geometry

Multifactor extensions should retain their actual dimension. Candidate coordinates include products of one-dimensional charts, projective or spherical charts with multiple phases, and manifold methods whose diffusion rank is explicit. Hörmander brackets and support must be analyzed rather than inferred from driver count. A single phase should be used only after a sufficient-statistic or invariant-manifold theorem proves losslessness.

One-phase coordinate implication decisions

claim	decision	evidence
exact lossless affine representation	supported	machine-precision reconstruction and proof
additive phase winds	refuted	phase domain has length π
Cayley is the original additive process	refuted	it is an invertible nonlinear transform
Cayley is circular Brownian motion	refuted	angular diffusion is state dependent
one driver creates rank-one terminal law in every system	refuted	hypocoelliptic control has full-rank covariance
complex embedding creates holomorphic derivatives	refuted	normal derivative depends on off-curve extension
phase coordinate reduces ordinary MC variance	refuted	identical streams give identical samples
phase coordinate lowers direct terminal-estimator work	refuted	same random dimension plus an arctangent/tangent round trip
compactification always improves PDE numerics	refuted	accuracy and conditioning depend on c and resolution
compatible multiplicative phase can wind losslessly	supported	logarithmic spiral reconstruction
geometry can preserve a transformed invariant	supported	exact Cayley map preserves unit modulus
new analytic solution for ABM	not claimed	exact phase kernel is the known Gaussian kernel under a diffeomorphism

Figure 22: Evidence-based decisions for the principal claims and negative controls.

13.2.5 Path-dependent claims

The terminal phase cannot encode a Brownian bridge. For Asian, barrier, and early-exercise claims, legitimate approaches include augmenting the Markov state with running integrals or extrema, simulating the full phase path, or using regression and dynamic programming (Longstaff and Schwartz 2001; Zhang 2003). The phase may still serve as one coordinate of that expanded state.

13.2.6 Beyond affine and log-affine rigidity

For a prescribed smooth radius $r(\theta)$ and phase diffusion, Eq. 33 constructs a state-dependent complex SDE. A broader classification could ask which drift-diffusion families preserve selected curves or admit global polar charts across zeros. Such work belongs to stochastic differential geometry and should not be inferred from the affine line alone.

13.3 Decision map

The decision map in Figure 22 enforces a conservative interpretation. Exactness, lossless reconstruction, the affine classification, and conjugate pricing are supported. Automatic variance reduction, ordinary circular Brownian behavior, added dimension, universal conditioning improvement, and new complex analyticity are not.

14 Limitations

The scope of the results is deliberately narrow.

1. **One real driver and constant affine coefficients.** The global iff theorem concerns $Z_0 + at + bW_t$. State-dependent coefficients, jumps, multiple Brownian drivers, and random coefficients require new support and regularity arguments.
2. **Genuine polar phase.** Pathological measurable encodings of higher-dimensional data into one real number are irrelevant to the smooth, geometric, Itô-compatible problem.
3. **Nonvanishing reference geometry.** The main normalization assumes $Z_0 \neq 0$ and the eligible line misses the origin. Charts across zeros need stopping times or multiple coordinate patches.
4. **No unlimited additive winding.** The additive phase interval has length π . Unlimited winding belongs to the compatible multiplicative spiral, which is a different process.
5. **Normal-model financial scope.** Arithmetic Brownian levels can become negative. The model is not a universal equity-price model, and the complex coordinate is not traded.
6. **Classical smooth pricing.** The PDE derivation assumes sufficient regularity. Viscosity, weak, or distributional formulations are needed for nonsmooth coefficients and payoffs, although conditional-expectation invariance remains immediate when integrable.
7. **Finite numerical audit.** Convergence slopes, residuals, and compactification errors are finite-resolution measurements. Only the algebraic identities are exact to mathematical precision.
8. **No universal computational advantage.** The heat-equation benchmark shows a resolution-conditioning trade-off for one problem. It does not establish superiority across payoffs, solvers, dimensions, or hardware.
9. **No dimensional or analytic creation.** A coordinate diffeomorphism cannot create independent noise, path history, market completeness, or an off-curve analytic extension.

15 Conclusion

For the nondegenerate affine complex Brownian process

$$Z_t = Z_0 + at + bW_t,$$

a lossless, time-independent genuine Euler phase exists exactly when

$$\frac{a}{b} \in \mathbb{R}, \quad \frac{b}{Z_0} \notin \mathbb{R}.$$

The first condition fixes the full temporal support to one affine line. The second makes the line's argument injective. With $a = \lambda b$ and $b/Z_0 = \rho e^{i\phi}$, the unique continuous branch has

$$Z_t = Z_0 \frac{\sin \phi}{\sin(\phi - \theta_t)} e^{i\theta_t}, \quad x_t = \frac{\sin \theta_t}{\rho \sin(\phi - \theta_t)}.$$

The canonical financial representative is

$$1 + iX_t/c = \sec \theta_t e^{i\theta_t}, \quad X_t = c \tan \theta_t.$$

Its phase is an autonomous diffusion with explicit generator, transition kernel, exact step, and inaccessible natural boundaries. The chart conjugates the Bachelier pricing problem, including values, deltas, gammas, and hedges. It can concentrate numerical resolution and preserve transformed invariants, but exact samplewise inversion leaves Monte Carlo estimators unchanged.

The rotational aspect of Euler form is therefore real but constrained. In the additive case, rotation and phase-dependent radius trace a straight line over one half-turn. Unlimited lossless winding is available only after moving to a compatible multiplicative logarithmic spiral. Complex notation clarifies tangents, normals,

and coefficient squares; Itô quadratic variation supplies the second-order correction. Neither substitutes for the other.

The useful outcome is not a claim that stochastic calculus has been reduced to complex algebra. It is a precise classification and a controlled coordinate framework: exact where the support geometry permits it, explicit about what financial quantities it preserves, and equally explicit about the dimensions, histories, and analytic structures it cannot create.

Appendix A: Additional proof details

A.1 Simultaneous-time formulation

The explicit construction in the sufficient case holds pathwise for every $t \geq 0$ on the event where W is continuous, which has probability one. For necessity, it is enough to assume the representation holds almost surely at each deterministic t in $[t_0, t_1]$. Define

$$H(t, \omega) = \mathbf{1}\{Z_t(\omega) \notin Z_0 C_r\}.$$

Assuming standard joint measurability, $\mathbb{E}[H(t, \cdot)] = 0$ for every t . Tonelli's theorem gives

$$\int_{t_0}^{t_1} \mathbb{E}[H(t, \cdot)] dt = 0,$$

so an independent uniform τ satisfies $\mathbb{P}(Z_\tau \in Z_0 C_r) = 1$. This is the step used in the randomized-time contradiction.

A.2 Determinant of the affine time-state map

Write $a = a_1 + ia_2$ and $b = b_1 + ib_2$. The real Jacobian of

$$(t, w) \mapsto Z_0 + at + bw$$

is

$$J = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}, \quad \det J = a_1 b_2 - a_2 b_1 = -\operatorname{Im}(a\bar{b}).$$

Thus $a/b \notin \mathbb{R}$ is exactly the condition for the randomized-time law to inherit a planar density.

A.3 Continuous-lift uniqueness

The curve $x \mapsto 1 + cx$ is continuous and nonvanishing in the eligible case. Any two continuous argument lifts differ by a continuous $2\pi\mathbb{Z}$ -valued function, hence by a constant multiple of 2π . Anchoring both at $\theta(0) = 0$ makes them equal. Strict monotonicity follows from Eq. 13, and eliminating x forces the radius in Eq. 14.

A.4 General moving-line fallback

At a fixed time t , write the normalized support as

$$\ell_t(x) = q_t + cx, \quad q_t = 1 + \alpha t.$$

When $\text{Im}(cq_t) \neq 0$, this line misses the origin, its angle is monotone, and the same line-polar calculation gives a radius $R(t, \theta)$. At times satisfying

$$\text{Im}(cq_t) = 0,$$

the line passes through the origin and no single lossless genuine polar phase represents its full Gaussian support. The explicit dependence on t is why this construction does not contradict Theorem 5.1.

Appendix B: General affine stochastic theory

For $x_t = \lambda t + W_t$, the inverse derivative is

$$x'(\theta) = \frac{\sin \phi}{\rho \sin^2(\phi - \theta)},$$

with sign determined by orientation. Therefore

$$g(\theta) = \frac{1}{x'(\theta)} = \frac{\rho \sin^2(\phi - \theta)}{\sin \phi}.$$

Differentiating $\theta(x)$ twice gives $\theta''(x) = g(\theta)g'(\theta)$, and Itô's formula yields Eq. 27.

The corresponding generator is

$$\mathcal{L}_\phi h = \left(\lambda g + \frac{1}{2} g g' \right) h' + \frac{1}{2} g^2 h''.$$

Under the substitution $y = x(\theta)$, this generator is conjugate to

$$\mathcal{L}_x f = \lambda f' + \frac{1}{2} f''.$$

The transition density Eq. 31, its normalization, and the Chapman–Kolmogorov identity follow from the same conjugacy. Both endpoints of I_ϕ map to infinite scalar states and are natural.

Appendix C: Financial derivative identities

C.1 Bachelier expectation

Under Eq. 41,

$$F_T = F_t + \sigma_N \sqrt{\tau} \xi, \quad \xi \sim N(0, 1).$$

For $s = \sigma_N \sqrt{\tau} > 0$,

$$\mathbb{E}[(F_T - K)^+ | F_t = F] = \int_{(K-F)/s}^{\infty} (F - K + sz) \varphi(z) dz.$$

Using

$$\int_a^{\infty} \varphi(z) dz = \Phi(-a), \quad \int_a^{\infty} z \varphi(z) dz = \varphi(a),$$

gives Eq. 51. Differentiation under the closed form yields Eq. 53.

C.2 Greek transformation

Since

$$\theta_x = \frac{\cos^2 \theta}{c},$$

the first derivative is

$$V_x = v_\theta \theta_x = \frac{\cos^2 \theta}{c} v_\theta.$$

Differentiating again,

$$\theta_{xx} = -\frac{2 \sin \theta \cos^3 \theta}{c^2},$$

and therefore

$$V_{xx} = v_{\theta\theta} \theta_x^2 + v_\theta \theta_{xx},$$

which is Eq. 55.

C.3 Generator cancellation

Substituting Eq. 48 and Eq. 49 into the phase operator gives

$$\begin{aligned} Av_\theta + \frac{1}{2}B^2v_{\theta\theta} &= \left(\frac{\mu_Q}{c} \cos^2 \theta - \frac{\sigma^2}{c^2} \sin \theta \cos^3 \theta \right) V_x c \sec^2 \theta \\ &\quad + \frac{\sigma^2}{2c^2} \cos^4 \theta (V_{xx} c^2 \sec^4 \theta + 2V_x c \tan \theta \sec^2 \theta) \\ &= \mu_Q V_x + \frac{1}{2} \sigma^2 V_{xx}. \end{aligned}$$

The two terms proportional to $\sigma^2 V_x \sin \theta / \cos \theta$ cancel exactly.

Appendix D: Reproducibility and artifact manifest

The computational supplement consists of:

- the deterministic builder `notebooks/build_one_phase_euler_finance.py`;
- the executed notebook `notebooks/one_phase_euler_finance.ipynb`;
- 22 vector figures in `figures/vector`;
- 22 raster companions in `figures/raster`; and
- 44 structured evidence files in `tables`.

The notebook records the following primary parameters:

Table 6: Principal computational parameters.

Experiment	Parameters
Nonlinear SDE	$\mu = 0.3$, $\sigma = 0.8$, $c = 1.25$, $T = 1$, 4,000 paths, 32–512 steps
Boundary stress	$c = 0.25$, $\sigma = 1$, 120,000 samples per step size
Distribution law	signed $\sigma = -0.9$ with independent PIT and quadrature checks
Monte Carlo	2^{18} samples, shared scalar/phase streams
Bachelier pricing	$c = 0.05$, $K = 0.025$, $\sigma_N = 0.0125$, $\tau = 1.25$, $D = 0.97$
Pricing PDE	2,869 interior phase-maturity evaluations
Pinned paths	28,000 Brownian bridges with 80 steps
Two-factor control	120,000 correlated Gaussian samples

The final notebook assertions require:

$$\begin{aligned} &\text{all 44 figure files present and nonempty,} \\ &\max |C_{\text{direct}} - C_{\text{phase}}| < 10^{-14}, \\ &\max |\Delta_{\text{direct}} - \Delta_{\text{phase}}| < 2 \times 10^{-6}, \\ &\max |\Gamma_{\text{direct}} - \Gamma_{\text{phase}}| < 2 \times 10^{-3}, \\ &\max |\text{pricing-PDE residual}| < 2 \times 10^{-4}, \end{aligned}$$

and rank two for the multifactor covariance control. The observed values were much smaller: 2.78×10^{-17} , 1.76×10^{-8} , 5.81×10^{-7} , and 8.12×10^{-10} , respectively.

All results reported in the manuscript are read from the executed evidence files. The notebook is intended to be rerun from a clean kernel; the vector and raster manifests allow missing or stale figures to be detected independently of the prose.

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